



Xger

Reference Manual

Version 2.49

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1 Overview

```
# Changelog

## 2.48
- **Changed**
  - `audioProfile` from 2.4 to 2.5
  - `audioStatus` from 2.13 to 2.14
  - `coderConfig` from 2.39 to 2.40
  - `coderService` from 2.37 to 2.38
  - `multiService` from 2.9 to 2.10
  - `multiServiceProfile` from 2.2 to 2.3
  - `serviceStatus` from 2.38 to 2.39
  - `st2110Status` from 1.3 to 1.4
  - `vancStatus` from 1.10 to 1.11

## 2.47
- **Added**
  - 'scte35LogApi' version 1.0
  - sdiRouterConfig version 1.1
- **Changed**
  - `vancStatus` from 1.9 to 1.10
  - `serviceStatus` from 2.37 to 2.38

## 2.46
- **Added**
  - `nmosStatusTypes` version 1.1
- **Changed**
  - `nmosConfig` from 1.0 to 1.1
  - `cardConfig` from 1.2 to 1.3
  - `videoProfile` from 2.8 to 2.9
  - `expectedInput` from 1.2 to 1.3
  - `multiServiceProfile` from 2.1 to 2.2
  - `cardAllocation` from 1.5 to 1.6
  - `serviceStatus` from 2.36 to 2.37
  - `videoStatus` from 2.11 to 2.12
  - `coderService` from 2.36 to 2.37
  - `multiService` from 2.8 to 2.9
  - `cardStatus` from 1.7 to 1.8

## 2.45
  - `multiService` from 2.7 to 2.8
  - `coderService` from 2.35 to 2.36
  - `tsSource` from 1.3 to 1.4
  - `serviceStatus` from 2.35 to 2.36
  - `vancStatus` from 1.8 to 1.9

## 2.44
- **Changed**
  - `multiService` from 2.6 to 2.7
  - `coderService` from 2.34 to 2.35
  - `tsSource` from 1.2 to 1.3
  - `ipStream` from 1.2 to 1.3
  - `st2110Types` from 1.0 to 1.2
  - `serviceStatus` from 2.34 to 2.35
  - `st2110Status` from 1.2 to 1.3
  - `ipConnection` from 1.4 to 1.5
  - `ipService` from 1.1 to 1.2
  - `rawAudio` from 1.0 to 1.1
  - `audioProfile` from 2.3 to 2.4
```

- `multiServiceProfile` from 2.0 to 2.1
- `audioStatus` from 2.12 to 2.13
- `vancStatus` from 1.7 to 1.8
- `vancProfile` from 1.4 to 1.5

2.43

- ****Added****
 - `logoInsertion` 1.0
- ****Changed****
 - `coderService` from 2.33 to 2.34
 - `multiService` from 2.5 to 2.6
 - `audioStatus` from 2.11 to 2.12
 - `sdiStatus` from 1.8 to 1.9
 - `serviceStatus` from 2.33 to 2.34

2.42

- ****Changed****
 - `serviceStatus` from 2.32 to 2.33
 - `vancStatus` from 1.6 to 1.7
 - `cardConfig` from 1.1 to 1.2
- ****Added****
 - `nmosConfig` 1.0
 - `nmosLabelConfig` 1.0

2.41

- ****Changed****
 - `vancProfile` from 1.3 to 1.4
 - `coderService` from 2.32 to 2.33
 - `vancStatus` from 1.5 to 1.6
 - `multiService` from 2.4 to 2.5
 - `serviceStatus` from 2.31 to 2.32

2.40

- ****Changed****
 - `coderService` from 2.31 to 2.32
 - `multiService` from 2.3 to 2.4
 - `ipInterface` from 1.4 to 1.5
 - `slotIpInterface` from 1.4 to 1.5
 - `linkModes` from 1.1 to 1.2

2.39

- ****Added****
 - `poisServerStatus` version 1.0
- ****Changed****
 - `poolCtrlAllocs` from 1.0 to 1.1
 - `lockStatus` from 1.2 to 1.3
 - `coderService` from 2.30 to 2.31
 - `serviceStatus` from 2.30 to 2.31
 - `multiService` from 2.2 to 2.3

2.38

- ****Added****
 - `esamConfig` version 1.0
- ****Changed****
 - `multiService` from 2.1 to 2.2
 - `poolConfig` from 1.1 to 1.2
 - `coderService` from 2.29 to 2.30

2.37

- ****Changed****
 - `coderService` from 2.28 to 2.29
 - `multiService` from 2.0 to 2.1
 - `dejitterTypes` from 1.0 to 1.1
 - `ipConnection` from 1.3 to 1.4
 - `ipService` from 1.0 to 1.1
 - `videoStatus` from 2.10 to 2.11
 - `serviceStatus` from 2.29 to 2.30
 - `psiTypes` from 1.5 to 1.6
 - `psiStatus` from 1.5 to 1.6
 - `inputRedundancyStatus` from 1.2 to 1.3

2.36

- ****Changed****
 - `coderService` from 2.27 to 2.28
 - `multiServiceProfile` from 1.10 to 2.0
 - `multiService` from 1.22 to 2.0
 - `ipConnection` from 1.2 to 1.3
 - `serviceStatus` from 2.28 to 2.29
 - `sdpUpload` from 1.0 to 1.1
- ****Added****
 - `ipStreamUtils` version 0.0
 - `ipService` version 1.0

2.35

- ****Changed****
 - `coderService` from 2.26 to 2.27
 - `multiService` from 1.21 to 1.22
 - `sdiStatus` from 1.7 to 1.8
 - `serviceStatus` from 2.27 to 2.28

2.34

- ****Changed****
 - `redundancyGroupStatus` from 1.1 to 1.2;

2.33

- ****Added****
 - `slateInsertion` 1.0
 - `imageUpload` 1.0
- ****Changed****
 - `multiService` from 1.20 to 1.21
 - `coderService` from 2.25 to 2.26

2 afd (1.0)

2.1 Type Reference

2.1.1 AfdCode

enum

UNKNOWN

RESERVED_1

BOX_16_9_TOP

BOX_14_9_TOP

BOX_16_9_CENTRE

RESERVED_5

RESERVED_6

RESERVED_7

SAME

_4_3_CENTRE

_16_9_CENTRE

_14_9_CENTRE

RESERVED_12

_4_3_PROTECT_14_9_CENTRE

_16_9_PROTECT_14_9_CENTRE

_16_9_PROTECT_4_3_CENTRE

3 alternateInputStatus (1.4)

3.1 Type Reference

3.1.1 AlternateInputStatus

- When alternative input is used, this struct holds the lists of IpStreamStatus of both input paths.
- It also holds the status of the alternative or seamless input configuration.
-

struct

path1	list of IpStreamStatus.IpStreamStatus Status of the IpStreams on path 1. •
path2	list of IpStreamStatus.IpStreamStatus Status of the IpStreams on path 2. •
seamlessType	list of SeamlessTypeStatus list of statuses of the detected type of seamless input.

3.1.2 NearSeamlessStatus

Status of a near seamless input.

struct

active	Path The currently active path. Only used for near seamless.
synchronized	bool True if the inputs are synchronized.

3.1.3 Path

The path of a near seamless input which a status describes.

enum

path1	
path2	

3.1.4 SeamlessTypeStatus

Status of a SMPTE 2022-7 Seamless input or a near seamless input.

variant

smp2022_7	Smp2022_7Status Status of a SMPTE 2022-7 Seamless input.
	NearSeamlessStatus

`nearSeamless`

Status of a near seamless input.

3.1.5 Smppte2022_7Status

Status of a SMPTE 2022-7 Seamless input.

struct

<code>id</code>	UUID The id of the IpStream this status represents.
<code>synchronized</code>	bool True if the inputs are synchronized.
<code>relativeDelay</code>	int Delay between path A and path B in milliseconds.
<code>margin</code>	int Difference between the delay buffer size and the current relative delay in milliseconds. It is what remains of the buffer.
<code>maxRelativeDelay</code>	int The size of the delay buffer in milliseconds.

4 aspectRatio (1.0)

4.1 Type Reference

4.1.1 AspectRatioStandard

enum

SMPTE_2016

WSS

VIDEO_INDEX

5 audioProfile (2.5)

5.1 Overview

```
# Changelog
## 2.5
- **Added**
  - followInputChannelMode as optional(bool) in `AudioProfileStruct`

## 2.4
- **Changed**
  - `rawAudio` from 1.0 to 1.1
```

5.2 Command Reference

5.2.1 GetAudioProfiles

- message **GetAudioProfiles.Request**
- message **GetAudioProfiles.Response**
- message **GetAudioProfiles.Failure**

5.2.2 SetAudioProfiles

- message **SetAudioProfiles.Request**
- message **SetAudioProfiles.Response**
- message **SetAudioProfiles.Failure**

5.2.3 DeleteAudioProfiles

- message **DeleteAudioProfiles.Request**
- message **DeleteAudioProfiles.Response**
- message **DeleteAudioProfiles.Failure**

5.3 Type Reference

5.3.1 AacContainer

enum

ADTS

LATM

5.3.2 AacLcParameters

struct

container

AacContainer

5.3.3 AudioProfile

AudioProfileStruct

5.3.4 AudioProfileStruct

struct

label	string
codec	codedAudio.AudioCodec
channelMode	rawAudio.ChannelMode
followInputChannelMode	optional bool
sampleRate	SampleRate
bitRate	int
cparams	CodecParameters

5.3.5 CodecParameters

variant

mpeg1l2	Mpeg1l2Parameters
aac	AacLcParameters
dd	DolbyDigitalPlusParameters

5.3.6 CompProfileAndDialNorm

struct

dialnorm	int
compProfile	DdppCompressionProfile

5.3.7 DdppAdConverterType

enum

Undefined
Standard
HDCD

5.3.8 DdppBitstreamMode

enum

Undefined
CompleteMain
MusicAndEffects
VisuallyImpaired
HearingImpaired
Dialogue
Commentary
Emergency
VoiceOver
Karaoke

5.3.9 DdppChannelModeParameters

variant

mono	DdppChannelModeParameters.mono
stereo	DdppStereoParameters
multi	DdppMultiChannelParameters

5.3.10 DdppChannelModeParameters.mono

empty **struct**

5.3.11 DdppCompressionProfile

struct

dynrng	DdppDrcProfile
compr	DdppDrcProfile

5.3.12 DdppDownmixLevel

enum

Undefined
Pos_3Db
Pos_1_5Db
Pos_0Db
Neg_1_5Db
Neg_3Db
Neg_4_5Db
Neg_6Db
NegInfDb

5.3.13 DdppDownmixMode

enum

Undefined
NotIndicated
LtRt
LoRo
ProLogicII

5.3.14 DdppDrcProfile

enum

Undefined
None
FilmStandard

FilmLight
MusicStandard
MusicLight
Speech

5.3.15 DdppEncExtMd

struct

extMetadataProgramNum	int
extMetadataRevMode	DdppExtMetadataReversionMode
extMetadataOverwriteMode	DdppExtMetadataOverwriteMode

5.3.16 DdppEncManualMd

struct

dialnorm	int
bsmod	DdppBitstreamMode
compProfile	DdppCompressionProfile
mixlevel	int
roomtyp	DdppRoomType
copyrightb	bool
origbs	bool
adconvtyp	DdppAdConverterType
evolutionFrameworkEnabled	bool
cmParams	DdppChannelModeParameters

5.3.17 DdppExtMetadataOverwriteMode

variant

none	DdppExtMetadataOverwriteMode.none
dialnorm	int
compProfile	DdppCompressionProfile
compProfileAndDialNorm	CompProfileAndDialNorm

5.3.18 DdppExtMetadataOverwriteMode.none

empty struct

5.3.19 DdppExtMetadataOverwriteModeEnum

enum

NONE
DIALNORM
COMPPROFILE
COMPPROFILEANDDIALNORM

5.3.20 DdppExtMetadataReversionMode

enum

Undefined
 LastUsedMetadata
 LastHostSettings

5.3.21 DdppHeadphoneMode

enum

Undefined
 NotIndicated
 Disabled
 Enabled

5.3.22 DdppMetadata

variant

encManualMd	DdppEncManualMd
encExtMd	DdppEncExtMd
transMd	DdppTransMd

5.3.23 DdppMetadataEnum

enum

DDPPENCMANUALMD
 DDPPENCEXTMD
 DDPPTRANSMD

5.3.24 DdppMultiChannelParameters

struct

lfelpfon	bool
sur90on	bool
suratton	bool
lorocmixlev	DdppDownmixLevel
lorosurmixlev	DdppDownmixLevel
ltrtcmixlev	DdppDownmixLevel
ltrtsurmixlev	DdppDownmixLevel
dmixmod	DdppDownmixMode
dsurexmod	DdppSurroundMode

5.3.25 DdppRoomType

enum

Undefined

NotIndicated

LargeRoom

SmallRoom

5.3.26 DdppStereoParameters

struct

dsurmod

DdppSurroundMode

5.3.27 DdppSurroundMode

enum

Undefined

NotIndicated

Disabled

Enabled

5.3.28 DdppTransMd

struct

dialnorm

int

evolutionFrameworkEnabled

bool

5.3.29 DeleteAudioProfiles.Failure

empty struct

5.3.30 DeleteAudioProfiles.Request

struct

ids

list of UUID

5.3.31 DeleteAudioProfiles.Response

empty struct

5.3.32 DolbyDigitalPlusParameters

struct

metadata

DdppMetadata

5.3.33 GetAudioProfiles.Failure

empty **struct**

5.3.34 GetAudioProfiles.Request

empty **struct**

5.3.35 GetAudioProfiles.Response

struct

data	map from UUID to AudioProfile
info	GetAudioProfiles.Response.info

5.3.36 GetAudioProfiles.Response.info

struct

totalEntries	int
--------------	------------

5.3.37 Mpeg12Parameters

empty **struct**

5.3.38 SampleBitDepth

enum

SBD_16
SBD_20
SBD_24

5.3.39 SampleRate

enum

SR_48kHz

5.3.40 SetAudioProfiles.Failure

empty **struct**

5.3.41 SetAudioProfiles.Request

struct

data	map from UUID to AudioProfile
------	--

5.3.42 SetAudioProfiles.Response

empty **struct**

6 audioStatus (2.14)

6.1 Overview

```
# Changelog
## 2.14
- **Changed**
  - `audioProfile` from 2.4 to 2.5

## 2.13
- **Changed**
  - `rawAudio` from 1.0 to 1.1
  - `audioProfile` from 2.3 to 2.4

## 2.12
- **Changed**
  - `embeddedChannel` to `embeddedChannels` in `BasicConfig`; now a list of all channels in use a

## 2.11
- **Changed**
  - `component` from 1.3 to 1.5
```

6.2 Type Reference

6.2.1 Audio

struct

channelId	int
uid	int
source	Source
output	Output
incorrectDspSpeed	optional bool

6.2.2 AudioSet

struct

main	Audio
backup	optional Audio
active	Input

6.2.3 AudioStatus

struct

audios	list of AudioSet
--------	------------------

6.2.4 BasicConfig

struct

codec	optional codedAudio.AudioCodec
componentType	optional component.ComponentType
sampleRate	optional audioProfile.SampleRate
bitRate	optional int
channelMode	optional rawAudio.ChannelMode
numChannels	optional int
aacContainer	optional audioProfile.AacContainer
embeddedChannels	optional list of int

6.2.5 CondensedAudio

struct

channelId	int
uid	int
source	CondensedAudio.source
output	CondensedAudio.output

6.2.6 CondensedAudio.output

struct

codec	optional codedAudio.AudioCodec
channelMode	optional rawAudio.ChannelMode
bitRate	optional int
ddpp	optional CondensedAudio.output.ddpp

6.2.7 CondensedAudio.source

struct

addBsiByte0	optional int
-------------	---------------------

6.2.8 CondensedAudioStatus

struct

audios	list of CondensedAudio
--------	-------------------------------

6.2.9 DdppAcMod

enum

_1_0
_1_1
_2_0
_2_1
_2_2
_3_0

`_3_1``_3_2``_3_3``_3_4`

6.2.10 DdppDec

struct

<code>outputchannelconfig</code>	<code>rawAudio.ChannelMode</code>
<code>drcmode</code>	<code>DdppDecDrcMode</code>
<code>drchighcut</code>	<code>DdppDecDrcScale</code>
<code>drclowboost</code>	<code>DdppDecDrcScale</code>
<code>stereodownmixmode</code>	<code>DdppDecStereoDownmixMode</code>

6.2.11 DdppDecDrcMode

enum

<code>Undefined</code>	
<code>Custom_0</code>	
<code>Line</code>	
<code>Rf</code>	
<code>portable_Neg_8_Db</code>	
<code>portable_Neg_11_Db</code>	
<code>portable_Neg_14_Db</code>	

6.2.12 DdppDecDrcScale

enum

<code>Undefined</code>	
<code>Min</code>	
<code>Max</code>	

6.2.13 DdppDecStereoDownmixMode

enum

<code>Undefined</code>	
<code>Auto</code>	
<code>LtRt</code>	
<code>LoRo</code>	

6.2.14 DdppEnc

struct

<code>dialnorm</code>	<code>int</code>
-----------------------	------------------

bsmod	audioProfile.DdppBitstreamMode
dynrng	audioProfile.DdppDrcProfile
compr	audioProfile.DdppDrcProfile
dsurmod	audioProfile.DdppSurroundMode
mixlevel	int
roomtyp	audioProfile.DdppRoomType
copyrightb	YesNo
origbs	YesNo
lorocmixlev	audioProfile.DdppDownmixLevel
lorosurmixlev	audioProfile.DdppDownmixLevel
ltrtcmixlev	audioProfile.DdppDownmixLevel
ltrtsurmixlev	audioProfile.DdppDownmixLevel
dmixmod	audioProfile.DdppDownmixMode
dsurexmod	audioProfile.DdppSurroundMode
dheadphonmod	audioProfile.DdppHeadphoneMode
adconvtyp	audioProfile.DdppAdConverterType
lfelpfon	OnOff
sur90on	OnOff
suratton	OnOff
evolutionFrameworkEnabled	OnOff

6.2.15 DdppMetadata

struct

acmod	DdppAcMod
lfeon	OnOff
datarate	optional int
samplerate	optional int
dialnorm	int
dialnorm2	optional int
bsmod	audioProfile.DdppBitstreamMode
dsurmod	audioProfile.DdppSurroundMode
mixlevel	optional int
roomtyp	optional audioProfile.DdppRoomType
copyrightb	YesNo
origbs	YesNo
cmixlev	optional audioProfile.DdppDownmixLevel
surmixlev	optional audioProfile.DdppDownmixLevel
lorocmixlev	audioProfile.DdppDownmixLevel
lorosurmixlev	audioProfile.DdppDownmixLevel
ltrtcmixlev	audioProfile.DdppDownmixLevel
ltrtsurmixlev	audioProfile.DdppDownmixLevel
dmixmod	audioProfile.DdppDownmixMode
dsurexmod	audioProfile.DdppSurroundMode
dheadphonmod	optional audioProfile.DdppHeadphoneMode
adconvtyp	audioProfile.DdppAdConverterType
adconvtyp2	optional audioProfile.DdppAdConverterType
addbsie	optional YesNo

addbsibyte0

optional int

6.2.16 DepMetadata

struct

frameRate	float
numberOfPrograms	int
programConfig	DolbyEProgramConfig
bitDepth	int
programSelect	optional int

6.2.17 DolbyDigitalPlusMetEvoStatus

enum

Nofound	
ErrorParse	
ErrorPcmhash	
ErrorSerialize	
valid	

6.2.18 DolbyEProgramConfig

enum

_51_2	
_51_1_1	
_4_4	
_4_2_2	
_4_2_1_1	
_4_1_1_1_1	
_2_2_2_2	
_2_2_2_1_1	
_2_2_1_1_1_1	
_2_1_1_1_1_1_1	
_1_1_1_1_1_1_1_1	
_51	
_4_2	
_4_1_1	
_2_2_2	
_2_2_1_1	
_2_1_1_1_1	
_1_1_1_1_1_1_1	
_4	
_2_2	
_2_1_1	
_1_1_1_1_1	
_71	

_71_Screen	
------------	--

6.2.19 EvolutionFrameworkMetadata

struct

evoPresent	YesNo
evoStatus	DolbyDigitalPlusMetEvoStatus

6.2.20 Input

enum

MAIN	
BACKUP	

6.2.21 LongTermLoudness

struct

loudnessLevel	optional float
maxTruePeak	optional float
gain	float

6.2.22 Loudness

struct

ltll	LongTermLoudness
stll	optional YesNo
peakLimiter	optional YesNo

6.2.23 OnOff

enum

Undefined	
On	
Off	

6.2.24 Output

struct

pid	optional int
basicConfig	BasicConfig
advancedConfig	optional DdppEnc
cannotEncode	bool
loudness	optional Loudness

6.2.25 Process

enum

ENCODE
TRANSCODE
DECODE
PASSTHROUGH

6.2.26 Source

struct

process	Process
pid	optional int
basicConfig	BasicConfig
advancedConfig	optional DdppDec
bitstreamMetadata	Source.bitstreamMetadata
cannotDecode	optional bool

6.2.27 Source.bitstreamMetadata

struct

dep	optional DepMetadata
ddpp	optional DdppMetadata
evo	optional EvolutionFrameworkMetadata

6.2.28 YesNo

enum

Undefined
Yes
No

7 cardAllocation (1.6)

7.1 Overview

Changelog:

1.6

- Changed
 - expectedInput from 1.2 to 1.3

1.5

- Changed
 - expectedInput from 1.1 to 1.2

7.2 Command Reference

7.2.1 GetCardAllocations

- message **GetCardAllocations.Request**
- message **GetCardAllocations.Response**

7.2.2 AssessAllocationPossibilities

- message **AssessAllocationPossibilities.Request**
- message **AssessAllocationPossibilities.Response**
- message **AssessAllocationPossibilities.Failure**

7.2.3 ReallocateAll

- message **ReallocateAll.Request**
- message **ReallocateAll.Response**
- message **ReallocateAll.Failure**

7.3 Type Reference

7.3.1 AssessAllocationPossibilities.Failure

empty **struct**

7.3.2 AssessAllocationPossibilities.Request

empty **struct**

7.3.3 AssessAllocationPossibilities.Response

struct

chipsRequired **int**

7.3.4 CardAllocation

struct

slot	optional int
inputInstance	optional int
outputInstance	optional int
enableDeinterlacer	optional bool
useCase	optional list of ResourceType
expectedInput	expectedInput.ExpectedInput

7.3.5 CardAllocationObj

CardAllocation

7.3.6 GetCardAllocations.Request

empty struct

7.3.7 GetCardAllocations.Response

struct

data	map from UUID to CardAllocationObj
info	GetCardAllocations.Response.info

7.3.8 GetCardAllocations.Response.info

struct

totalEntries	int
--------------	-----

7.3.9 ReallocateAll.Failure

empty struct

7.3.10 ReallocateAll.Request

empty struct

7.3.11 ReallocateAll.Response

struct

success	bool
---------	------

7.3.12 ResourceType

enum

NOT_USE	
RES_4K2K	
FULL_HD	
HD	
SD	

8 cardConfig (1.3)

8.1 Overview

Changelog

1.3

- **Changed**
 - nmosConfig from 1.0 to 1.1

1.2

- **Added**
 - nmosConfig 1.0
- **Removed**
 - nmosRegistryConfig 1.0

8.2 Command Reference

8.2.1 GetCardConfig

- message **GetCardConfig.Request**
- message **GetCardConfig.Response**
- message **GetCardConfig.Failure**

8.2.2 SetCardConfig

- message **SetCardConfig.Request**
- message **SetCardConfig.Response**
- message **SetCardConfig.Failure**

8.3 Type Reference

8.3.1 CardConfig

CardConfigStruct

8.3.2 CardConfigStruct

Card configuration parameters

struct

timeReference

timeMode.TimeMode

Time reference (PTP/NTP)

nmos

optional nmosConfig.NmosConfig

Parameters for accessing NMOS registry and Label Templates

8.3.3 GetCardConfig.Failure

empty **struct**

8.3.4 GetCardConfig.Request

empty **struct**

8.3.5 GetCardConfig.Response

struct

config **CardConfig**

8.3.6 SetCardConfig.Failure

empty **struct**

8.3.7 SetCardConfig.Request

struct

config **CardConfig**

8.3.8 SetCardConfig.Response

empty **struct**

9 cardStatus (1.8)

9.1 Overview

Changelog:

1.8

- **Changed**
 - Moved NmosStatus to nmosStatusTypes version 1.1

1.7

- **Added**
 - nodeApi map of interfaces to node API URIs
 - NmosStatus structure defined, and added to cardStatus structure.
 - NmosRegistryStatus put into NmosStatus.
- **Changed**
 - LinkModes from 1.2 to 1.3

1.6

- **Changed**
 - linkModes from 1.1 to 1.2

1.5

- **Changed**
 - sfpStatus from 1.4 to 1.5

9.2 Command Reference

9.2.1 GetCardStatus

- message **GetCardStatus.Request**
- message **GetCardStatus.Response**
- message **GetCardStatus.Failure**

9.3 Type Reference

9.3.1 CardStatus

struct

physicalPortStatus	optional PhysicalPortStatus
ptpLock	optional bool
nmosStatus	optional nmosStatusTypes.NmosStatus
qsfpStatus	optional map from physicalIpPort.PortName to sfpStatus.SfpStatus

9.3.2 DataRate

bigint

9.3.3 GetCardStatus.Failure

empty **struct**

9.3.4 GetCardStatus.Request

GetCardStatusRequest

9.3.5 GetCardStatus.Response

CardStatus

9.3.6 GetCardStatusRequest

struct

slot **int**

9.3.7 PhysicalPortStatus

struct

linkSpeed	map	from	physicalIpPort.PortName	to
			linkModes.IpLinkSpeed	
dataRate	map	from	physicalIpPort.PortName	to DataRate
lldpNeighbor	map	from	physicalIpPort.PortName	to
			lldp.LldpNeighborStruct	

10 codedAudio (1.0)

10.1 Type Reference

10.1.1 AudioCodec

enum

PCM

MPEG1L2

MPEG1L3

AAC_LC

AAC_HEv1

AAC_HEv2

AC3

EAC3

DOLBY_E

10.1.2 Hack

struct

profile

int

11 coderService (2.38)

11.1 Overview

```
# Changelog
## 2.38
- **Changed**
  - `audioProfile` from 2.4 to 2.5

## 2.37
- **Added**
  - New optional `pidSelection` member to `VancComponent`.
  - New list `pidList` member to `VancComponent`.
- **Changed**
  - `videoProfile` from 1.8 to 1.9

## 2.36
- **Changed**
  - `tsSource` from 1.3 to 1.4

## 2.35
- **Added**
  - `LR_CLfe` to `ChannelMapping` to match 3.1 channel mode use-case.
  - `identifier` optional field added to VancComponent struct, to be able to reference individual
- **Changed**
  - `tsSource` from 1.2 to 1.3
  - `ipStream` from 1.2 to 1.3
  - `ipService` from 1.1 to 1.2
  - `audioProfile` from 2.3 to 2.4
  - `vancProfile` from 1.4 to 1.5

## 2.34
- **Added**
  - `logoInsertion` 1.0
- **Changed**
  - EmbeddedAudio to use EmbeddedAudioChannels to enable custom source channels on encoders.

## 2.33
- **Changed**
  - `vancProfile` from 1.3 to 1.4
  - `EmbeddedAudio` `codec` made optional. Combine `codec` as empty optional with `passthrough`

## 2.32
- **Added**
  - `s302mPayload` to `TsAudioSource` in order to specify the content of a s302m PID

## 2.31
- **Changed**
  - `poolCtrlAllocs` from 1.0 to 1.1

## 2.30
- **Added**
  - `scte35Config` 1.0

- **Changed**
  - `idrOnScte35` optional bool changed to optional `Scte35Config`

## 2.29
- **Added**
```



```

- Optional list of `AudioComponentPattern` added to `EmbeddedAudio` to be used by ABR JPEG XS
- **Changed**
  - `ipService` from 1.0 to 1.1
  - Moved the `ColorComponent` type to its own file.
  - `ipConnection` removed as it is no longer in use here.

## 2.28
- **Added**
  - Added ipService import which has the IpService type which supports input profile redundancy.
- **Changed**
  - `VideoOut` edited to fit the new input redundancy
  - `ipConnection` from 1.2 to 1.3

## 2.27
- **Added**
  - `latencyAdjustment` field added to AudioComponent for forwarding audio earlier than video.

## 2.26
- **Added**
  - `slateInsertion` 1.0
  - slateInsertion field added to CoderService for configuration of the Slate Insertion feature.

## 2.25
- **Changed**
  - `component` from 1.3 to 1.5
  - `tsSource` from 1.1 to 1.2

## 2.24
- **Changed**
  - `testGenComponent` from 1.0 to 1.1
  - `testGeneratorProfile` from 1.0 to 1.1

## 2.23
- **Added**
  - `sdiConnection` 1.0
  - `tsDestination` 1.0
  - `testGenComponent` 1.0
- **Removed**
  - `SdiConnection` struct, moved to a separate file
  - `SdiConnectors` enum, moved to a separate file
  - `TestGenComponent` struct, moved to a separate file
  - `TsDestination` struct, moved to a separate file

```

11.2 Command Reference

11.2.1 GetCoderServices

- message **GetCoderServices.Request**
- message **GetCoderServices.Response**
- message **GetCoderServices.Failure**

11.2.2 SetCoderServices

- message **SetCoderServices.Request**
- message **SetCoderServices.Response**
- message **SetCoderServices.Failure**

11.2.3 DeleteCoderServices

- message **DeleteCoderServices.Request**
- message **DeleteCoderServices.Response**
- message **DeleteCoderServices.Failure**

11.3 Type Reference

11.3.1 AudioComponent

struct

uid	int
source	AudioSource
numAuPerPes	optional int
lipSyncAdjustment	optional int
latencyAdjustment	optional int
levelAdjustment	optional int
loudness	optional Loudness
passthrough	bool
profile	AudioComponent.profile
destination	AudioOut

11.3.2 AudioComponent.profile

struct

id	optional UUID
cpy	optional audioProfile.AudioProfileStruct

11.3.3 AudioComponentPattern

struct

language	string
component	optional component.ComponentType
pid	optional int

11.3.4 AudioComponentSet

struct

main	AudioComponent
backup	optional AudioComponent

11.3.5 AudioOut

variant

embedded	EmbeddedAudio
----------	----------------------

ts	TsAudio
----	----------------

11.3.6 AudioSource

variant

embedded	EmbeddedAudio
ts	TsAudioSource

11.3.7 AudioType

enum

UNDEFINED	
CLEAN_EFFECTS	
HEARING_IMPAIRED	
VISUALLY_IMPAIRED	

11.3.8 ChannelMapping

enum

C	
LR	
LR_CLfe	
LR_CLfe_LsRs	
LR_LsRs_CLfe	
LC_RLs_RsLfe	
LLs_CRs_RLfe	
LC_RsR_LsLfe	
CL_RLs_RsLfe	
LR_CLfe_LsRs_LbRb	
LR_LsRs_CLfe_LbRb	
LC_RLs_RsLfe_LbRb	
LLs_CRs_RLfe_LbRb	
LC_RsR_LsLfe_LbRb	
CL_RLs_RsLfe_LbRb	

11.3.9 CodeService

Configuration for an encoder or decoder service.

struct

label	string A label for the service.
enabled	bool True if the service should be enabled.
	int

slot	Which slot the service should run on (not applicable for Transcoders).
coderType	CoderType The type of coder the service is for.
video	VideoComponent Configuration of the input and output video for the coder chip.
audios	list of AudioComponentSet Configuration for the audio components.
output	OutputType The type of output that the service will produce.
color	optional colorComponent.ColorComponent Configuration for the color of the video.
vancs	list of VancComponent Configuration for the vanc components.
signalLoss	testGenerator.SignalLoss Configuration for the output when input signal is lost.
lock	LockComponent Configuration for how the service should be locked.
passthroughComponents	optional bool Set to true if all components should be passthrough. Only valid for transcoder.
additionalLatency	optional int Additional latency to be added to the entire service. Unit is 100ms. The optional value should be unset for products which do not support this feature.
testGenerator	optional testGenComponent.TestGenComponent Contains optional structure with link to a test generator profile.
scte35	optional scte35Config.Scte35Config
autoSelectUniqueAudioSourcePid	optional bool Automatically select unique PIDs for audio components. PIDs are selected in order of increasing numerical PID value from the PMT such that the first audio component uses the lowest PID, second audio component the next to lowest PID and so forth. PMT updates with changes to PID lineup will trigger a reconfiguration of all audio components. If the number of audio components exceed the number of audio PIDs in the PMT the extra audio components will not run. Only valid for decoders.
slateInsertion	optional slateInsertion.SlateInsertion Enable the slate insertion feature. Only supported on encoder.
logoInsertion	optional logoInsertion.LogoInsertion Contains optional structure with configuration for logo insertion.

11.3.10 CoderServiceObj

CoderService

11.3.11 CoderType

The types of coders that a config can be made for.

enum

```
ENCODER
TRANSCODER
DECODER
```

11.3.12 DdppDecDrcMode

enum

```
NONE
LINE
RF
```

11.3.13 DdppDecoder

DdppDecoder.

```
@param ddppDecDrcMode    Selects the mode for Dynamic Range Compression.
@param stereoDownmixMode Selects the mode for downmixing.
```

struct

```
ddppDecDrcMode          DdppDecDrcMode
stereoDownmixMode       StereoDownmixMode
```

11.3.14 DeleteCoderServices.Failure

empty **struct**

11.3.15 DeleteCoderServices.Request

struct

```
ids                      list of UUID
```

11.3.16 DeleteCoderServices.Response

empty **struct**

11.3.17 DolbyAudio

Dolby Audio.

```
@param dolbyEProgNum Selects a Dolby E program as source for the decoder and transcoder.
@param ddppdec         Specifies configuration for Dolby Digital and Dolby Digital Plus decoding.
```

struct

dolbyEProgNum	optional int
ddppdec	optional DdppDecoder

11.3.18 EmbeddedAudio

Configuration for embedded audio.

struct

codec	optional codedAudio.AudioCodec Audio codec
channels	EmbeddedAudioChannels Embedded audio content
dolbyEProgNum	optional int Dolby E program number
channelMode	optional rawAudio.ChannelMode Channel mode – mono, stereo, 5.1, etc.
channelMapping	optional ChannelMapping Dictates how embedded audio channels are interpreted with respect to order.
audioEssenceId	optional UUID Audio essence UUID
componentPriorityList	optional list of AudioComponentPattern Optional prioritized list of input components from a transport stream.

11.3.19 EmbeddedAudioChannels

States where embedded audio content should be read from or written to in the SDI signal.

struct

channels	list of int The source or destination channel(s) for the audio content
isCustom	bool States whether a user specified list deviates from the standard set, i.e. non-adjacent channels.

11.3.20 Genlock

struct

offset	int
--------	------------

11.3.21 GetCoderServices.Failure

empty **struct**

11.3.22 GetCoderServices.Request

empty **struct**

11.3.23 GetCoderServices.Response

struct

data	map from UUID to CoderServiceObj
info	GetCoderServices.Response.info

11.3.24 GetCoderServices.Response.info

struct

totalEntries	int
--------------	-----

11.3.25 IpOut

string

11.3.26 LockComponent

variant

source	bool
genlock	Genlock
videoAlignment	VideoAlignment

11.3.27 LockComponentEnum

enum

SOURCE
GENLOCK
VIDEOALIGNMENT

11.3.28 Loudness

struct

longTermLoudnessTargetLevel	int
shortTermDeviation	optional int
peakLimit	optional int

11.3.29 OutputType

The types of output that a service can produce.

variant

ts	tsDestination.TsDestination
sdi	SdiOut
ip	IpOut

11.3.30 PidSelection

enum

ALL
ANY
LIST

11.3.31 SdiOut

empty **struct**

11.3.32 SetCoderServices.Failure

empty **struct**

11.3.33 SetCoderServices.Request

struct

data	map from UUID to CoderServiceObj
------	--

11.3.34 SetCoderServices.Response

empty **struct**

11.3.35 StereoDownmixMode

enum

AUTO
LTRT
LORO

11.3.36 TsAudio

struct

language	string
pid	int
audioType	AudioType

11.3.37 TsAudioSource

Transport Stream Audio Source.

@param componentPriorityList A list of AudioComponentPattern objects describing audio components
 @param dolbyAudio Specifies decode configuration for Dolby Digital, Dolby Digital Plus
 @param s302mPayload Codec present in the s302m payload. Leave empty for transparent pas

struct

componentPriorityList	list of AudioComponentPattern
dolbyAudio	DolbyAudio
s302mPayload	optional codedAudio.AudioCodec

11.3.38 TsService

struct

pid	int
-----	-----

11.3.39 VancComponent

Configuration of VANC component

struct

uid	int Unique identifier for a component.
profile	VancComponent.profile Reference and copy to a VANC profile.
pidSelection	optional PidSelection Optional options for which PIDs to be selected
pidList	list of int List of PIDs to accompany PidSelection.LIST.
pid	int PID.
vancParams	vancProfile.VancParams Configuration of the type of ancillary data.
identifier	optional int Optional int to reference individual components.

11.3.40 VancComponent.profile

struct

id	UUID
cpy	optional vancProfile.VancProfileStruct

11.3.41 VideoAlignment

struct

index	int
alignmentMaster	optional poolCtrlAllocs.SlotFlow

11.3.42 VideoComponent

Video component of the coder service.

struct

source	VideoSource The input video
profile	VideoComponent.profile Video profile used by the coder service
destination	VideoOut The output video
timecodeSource	optional timecodeConfig.TimecodeSource Source of the timecode to be used for SEI messages (only set for encoders, but not for JPEG 2000 encoder)

11.3.43 VideoComponent.profile

struct

id	UUID
cpy	optional videoProfile.VideoProfileStruct

11.3.44 VideoOut

variant

sdi	sdiConnection.SdiConnection
ts	TsService
ip	ipService.IpService

11.3.45 VideoSource

```
Configuration type for the video source.
@param sdi configuration for SDI video source.
@param ts configuration for TS video source.
@param ip configuration for IP video source.
```

variant

sdi	sdiConnection.SdiConnection
ts	tsSource.TsSource
ip	ipService.IpService

12 coderTypes (1.0)

12.1 Type Reference

12.1.1 Range

struct

min	int
max	int

13 colorComponent (1.0)

13.1 Overview

```
# Changelog

## 1.0
- **Added**
  - Moved the ColorComponent type from coderService
```

13.2 Type Reference

13.2.1 ColorComponent

Configuration for a single color profile.

struct

profile	ColorComponent.profile The color profile with a UUID and a copy of the color profile struct
---------	---

13.2.2 ColorComponent.profile

struct

id	UUID
cpy	optional colorProfile.ColorProfileStruct

14 colorProfile (2.1)

14.1 Command Reference

14.1.1 GetColorProfiles

- message **GetColorProfiles.Request**
- message **GetColorProfiles.Response**
- message **GetColorProfiles.Failure**

14.1.2 SetColorProfiles

- message **SetColorProfiles.Request**
- message **SetColorProfiles.Response**
- message **SetColorProfiles.Failure**

14.1.3 DeleteColorProfiles

- message **DeleteColorProfiles.Request**
- message **DeleteColorProfiles.Response**
- message **DeleteColorProfiles.Failure**

14.2 Type Reference

14.2.1 ColorProfile

ColorProfileStruct

14.2.2 ColorProfileStruct

struct

label	string
colorType	ColorType
vui	VuiParameters

14.2.3 ColorSpace

enum

UNDEFINED
BT_709
BT_2020

14.2.4 ColorType

enum

OFF
HLG_10

HDR_10	
PQ_10	
SDR	
SDR_WCG	
HLG_10_ATF	
Manual	

14.2.5 DeleteColorProfiles.Failure

empty **struct**

14.2.6 DeleteColorProfiles.Request

struct

ids	list of UUID
-----	--------------

14.2.7 DeleteColorProfiles.Response

empty **struct**

14.2.8 GetColorProfiles.Failure

empty **struct**

14.2.9 GetColorProfiles.Request

empty **struct**

14.2.10 GetColorProfiles.Response

struct

data	map from UUID to ColorProfile
info	GetColorProfiles.Response.info

14.2.11 GetColorProfiles.Response.info

struct

totalEntries	int
--------------	-----

14.2.12 SetColorProfiles.Failure

empty **struct**

14.2.13 SetColorProfiles.Request

struct

data **map** from **UUID** to **ColorProfile**

14.2.14 SetColorProfiles.Response

empty **struct**

14.2.15 TransferFunc

enum

UNDEFINED

BT_709

BT_2010_SDR

BT_2011_PQ

BT_2011_HLG

14.2.16 VuiParameters

struct

colorSpace **ColorSpace**

transfer **TransferFunc**

matrix **ColorSpace**

15 commonTypes (1.0)

15.1 Type Reference

15.1.1 NmosVersionNumber

Generic version number type for NMOS APIs

enum

V1_2

V1_3

16 component (1.5)

16.1 Overview

Changelog:

1.5

- Added
 - SCTE27_SUB to ComponentType enum and respective mappings.

16.2 Type Reference

16.2.1 AnyComponent

empty **struct**

16.2.2 ComponentGroup

ComponentGroup: Grouping of ComponentType, used for filtering

enum

UNDEFINED
AUDIO
VIDEO
SUBTITLE
DATA

16.2.3 ComponentLanguage

string

16.2.4 ComponentSelector

Component selector.

@param pid	Selects a component using a specific (source) PID.
@param language	Selects a component by a three character language string.
@param type	Selects a component by its type.
@param group	Selects a component by group membership.
@param any	Selects any component.

variant

pid	Pid
language	ComponentLanguage
componentType	ComponentType
componentGroup	ComponentGroup
any	AnyComponent

16.2.5 ComponentType

A ComponentType is typically identified by a stream type (see ISO 13818-1 Table 2-29), often distinguished with another header field such as a descriptor tag.

enum

MPEG_VIDEO

MPEG_AUDIO

TTX

DVBSUB

ECM

PCR

PRIVATE

AC_3

H264

VBI

AAC

VC_1

SCS_EMM

DPI_CUE

EAC3

AIT

HBBTV_CAROUSEL

DATA_CAROUSEL

H265

S302M

JPEG2000

S2038

DVBTTMLSUB

JPEG_XS_VIDEO

SCTE27_SUB

16.2.6 Pid

int

16.2.7 SubtitleFormat

enum

DVB

DVB_HARD_OF_HEARING

EBU

EBU_HARD_OF_HEARING

17 de jitterTypes (1.1)

17.1 Overview

Changelog:

1.1

- Added
 - PCR value added to De jitterType enum
 - PCR added to De jitterType2str

17.2 Type Reference

17.2.1 De jitterType

enum

OFF

RTP_2022_6

PCR

18 esamConfig (1.0)

18.1 Type Reference

18.1.1 EsamConfig

struct

poisUri **string**

18.1.2 EsamServiceConfig

Multiservice ESAM configuration.

struct

acquisitionPointId **string**
The acquisition point ID that will be sent to the POIS server.

19 esamStatus (1.0)

19.1 Type Reference

19.1.1 EsamStatus

ESAM status.

struct

poisConnectionStatus

PoisConnectionStatus

An enum to represent the connection status with the POIS server.

19.1.2 PoisConnectionStatus

UNKNOWN means no communication with the server yet

enum

CONNECTED

NO_CONNECTION

UNKNOWN

20 expectedInput (1.3)

20.1 Overview

Changelog

1.3

- **Changed**
- `videoProfile` from 2.8 to 2.9

1.2

- **Added**
 - Added ExpectedTsInput struct
 - Added ExpectedRawInput variant
 - New fields in ExpectedInput:
 - ts: ExpectedTsInput,
 - raw: ExpectedRawInput
- **Changed**
 - ExpectedInput changed from a struct to a variant of either ExpectedTsInput or ExpectedRawInput

20.2 Type Reference

20.2.1 ExpectedInput

Expected input format.

variant

ts	ExpectedTsInput the expected input format for a TS video source.
raw	ExpectedRawInput the expected input format for a RAW video source.

20.2.2 ExpectedRawInput

Expected raw input.

struct

hResolution	rawVideo.HorizontalResolution the expected horizontal resolution of the input.
vResolution	rawVideo.VerticalResolution the expected vertical resolution of the input.
scan	videoProfile.ScanningMode the expected scanning mode of the input.
fps	rawVideo.FrameRate the expected frame rate per second of the input.

20.2.3 ExpectedTsInput

Expected ts input.

struct

interlaced	bool Enables handling of interlaced inputs. If set it may consume more resources.
codec	optional videoProfile.VideoCodec Empty optional represents Any codec (in set MPEG2 AVC HEVC)
allow4k	bool Enables handling of 4k inputs. If set it will consume more resources.
allow10bit	bool Enables handling of 10bit inputs. If set it will consume more resources.

21 firewallTypes (1.0)

21.1 Type Reference

21.1.1 IPAddress

string

21.1.2 PaginatedRequest

Instructs the server to return only a subset of the data found on the server

@param pageSize number of entries returned per request

@param pageNumber what page of the set to returned, determined by the slice [pageSize*pageNumber

struct

pageSize	int
pageNumber	int

21.1.3 PaginationInfo

struct

totalEntries	int
--------------	-----

the total number of entries on the server, before matching

21.1.4 SocketAddress

Tuple with IP address and port

struct

address	IPAddress
	IP address, either IPv4 or IPv6
port	int

21.1.5 UUID

string

22 imageUpload (1.0)

22.1 Overview

```
# Changelog

## 1.0
- **Added**
  - First version of module
```

22.2 Command Reference

22.2.1 GetImages

- message **GetImages.Request**
- message **GetImages.Response**

22.2.2 SetImage

Must be sent after an image has been uploaded. It will make sure that image is stored permanently and is available for selection.

- message **SetImage.Request**
- message **SetImage.Response**

22.2.3 DeleteImage

- message **DeleteImage.Request**
- message **DeleteImage.Response**

22.3 Type Reference

22.3.1 DeleteImage.Request

struct

id **string**

22.3.2 DeleteImage.Response

empty **struct**

22.3.3 GetImages.Request

empty **struct**

22.3.4 GetImages.Response

struct

images **list of ImageInfo**

22.3.5 ImageInfo

Information about an image.

struct

id	string a unique identifier for the image.
fileName	string the image file name.

22.3.6 SetImage.Request

struct

fileName	string
----------	---------------

22.3.7 SetImage.Response

struct

fileId	string A unique ID for the uploaded image. It is important to be aware that the ID is based on the content of the file. So if identical files (with different file names) are uploaded they will overwrite eachother.
--------	---

23 inputRedundancyStatus (1.3)

23.1 Overview

```
# Changelog

## 1.3
- **Changed**
  - `psiStatus` from 1.5 to 1.6

## 1.2
- **Changed**
  - `psiStatus` from 1.4 to 1.5
```

23.2 Type Reference

23.2.1 HotStandbyStatus

Hot Standby status for input redundancy.

struct

identicalSourcesStatus **optional** **IdenticalSourcesStatus**

23.2.2 IdenticalSourcesStatus

Identical sources status for input redundancy.

struct

measuredOffset **int**
Offset between two identical sources in ms.

23.2.3 SourceStatus

Source status for input redundancy.

struct

components	psiStatus.PsiStatus Components present in the given source.
present	bool True if the source is present, false otherwise.
active	bool True if the source is the active one in a input redundancy set-up.

23.2.4 TsInputRedundancyStatus

TS input redundancy status.

struct

sourceStatus	map from streamSource.StreamSource to SourceStatus Status for the given stream source.
hotStandbyStatus	HotStandbyStatus

24 ipConnection (1.5)

24.1 Overview

Changelog:

1.5

- Changed
 - ipStream from 1.2 to 1.3

1.4

- Changed
 - deJitterTypes from 1.0 to 1.1

1.3

- Added
 - MPEG_TS value added to IpStandard enum

1.2

- Added
 - nmosEnable field added to IpConnection, with a default value of false

24.2 Command Reference

24.2.1 GetIpConnections

- message **GetIpConnections.Request**
- message **GetIpConnections.Response**
- message **GetIpConnections.Failure**

24.2.2 SetIpConnections

- message **SetIpConnections.Request**
- message **SetIpConnections.Response**
- message **SetIpConnections.Failure**

24.2.3 DeletelpConnections

- message **DeletelpConnections.Request**
- message **DeletelpConnections.Response**
- message **DeletelpConnections.Failure**

24.3 Type Reference

24.3.1 DeJitter

struct

mode	deJitterTypes.DeJitterType
bufferSize	int

24.3.2 DeleteIpConnections.Failure

empty **struct**

24.3.3 DeleteIpConnections.Request

struct

ids **list of UUID**

24.3.4 DeleteIpConnections.Response

empty **struct**

24.3.5 GetIpConnections.Failure

empty **struct**

24.3.6 GetIpConnections.Request

empty **struct**

24.3.7 GetIpConnections.Response

struct

data	map from UUID to IpConnectionObj
info	GetIpConnections.Response.info

24.3.8 GetIpConnections.Response.info

struct

totalEntries **int**

24.3.9 IpConnection

Group of settings for the IP inputs/outputs of a single service.

struct

label	string Label for the entire Group.
connection	ipStream.Connection Connections settings like addresses, ports, etc.
de jitter	optional De jitter De jitter for the input of IP 2110 and IP 2022 encoders.
standard	IpStandard SMPTE standard that is used for this group.

bool

nmosEnable

for allowing NMOS system to update the input/output profiles

24.3.10 IpConnectionObj

IpConnection

24.3.11 IpStandard

enum

SMPTE_2022

SMPTE_2110

MPEG_TS

24.3.12 SetIpConnections.Failure

empty struct

24.3.13 SetIpConnections.Request

struct

data

map from UUID to IpConnectionObj

24.3.14 SetIpConnections.Response

empty struct

25 ipInterface (1.6)

25.1 Overview

```
# Changelog

## 1.6
- **Changed**
`linkModes` import from 1.2 to 1.3

## 1.5
- **Added**
`ipFecMode` added to IpInterfaceStruct

## 1.4
- **Changed**
`lldp` import from 1.0 to 1.1
```

25.2 Command Reference

25.2.1 GetIpInterfaces

- message **GetIpInterfaces.Request**
- message **GetIpInterfaces.Response**
- message **GetIpInterfaces.Failure**

25.2.2 SetIpInterfaces

Interface uuids are deterministically calculated The uuids used in the RPC Request are ignored

- message **SetIpInterfaces.Request**
- message **SetIpInterfaces.Response**
- message **SetIpInterfaces.Failure**

25.2.3 DeleteIpInterfaces

- message **DeleteIpInterfaces.Request**
- message **DeleteIpInterfaces.Response**
- message **DeleteIpInterfaces.Failure**

25.3 Type Reference

25.3.1 DeleteIpInterfaces.Failure

empty **struct**

25.3.2 DeleteIpInterfaces.Request

struct

ids **list of UUID**

25.3.3 DeleteIpInterfaces.Response

empty **struct**

25.3.4 GetIpInterfaces.Failure

empty **struct**

25.3.5 GetIpInterfaces.Request

empty **struct**

25.3.6 GetIpInterfaces.Response

struct

data **map** from **UUID** to **IpInterface**

25.3.7 IpInterface

IpInterfaceStruct

25.3.8 IpInterfaceConfig

struct

ipAddress	firewallTypes.IpAddress
gateway	firewallTypes.IpAddress
netmask	int

25.3.9 IpInterfaceStruct

struct

ipLink	linkModes.IpLinkMode
ipFecMode	optional linkModes.IpFecMode
interfaces	IpInterfaceVariant

25.3.10 IpInterfaceVariant

variant

single	LogicalInterface
quad	IpInterfaceVariant.quad

25.3.11 IpInterfaceVariant.quad

struct

first **LogicalInterface**

second	LogicalInterface
third	LogicalInterface
fourth	LogicalInterface

25.3.12 LogicalInterface

struct

label	string
enabled	bool
physicalPort	physicalIpPort.PortName
ipv4	optional IpInterfaceConfig
ipv6	optional IpInterfaceConfig
lldpMode	optional Ildp.LldpMode

25.3.13 SetIpInterfaces.Failure

empty struct

25.3.14 SetIpInterfaces.Request

struct

data	map from UUID to IpInterface
------	------------------------------

25.3.15 SetIpInterfaces.Response

empty struct

26 ipService (1.2)

26.1 Overview

```
# Changelog
## 1.2
- **Changed**
  - `ipConnection` from 1.4 to 1.5
  - `ipStream` from 1.2 to 1.3

## 1.1
- **Changed**
  - `ipConnection` from 1.3 to 1.4

## 1.0
- **Added**
  - New type `IpServices` added, containing multiple IP profiles and switching mode in order to
  - Moved the IpProfile type from coderService
```

26.2 Type Reference

26.2.1 IpProfile

Configuration for a single IP video source.

struct

id	UUID UUID of the IpConnection profile containing the configuration.
cpy	optional ipConnection.IpConnection A copy of the content of the IpConnection profile.

26.2.2 IpService

Configuration for one or more IP video sources.

struct

profiles	list of IpProfile
profileRedundancy	optional ipStream.NearSeamless
sdiType	sdiType.SdiType

27 ipStatus (1.7)

27.1 Overview

Changelog:

1.7

- Changed
 - sfpStatus from 1.4 to 1.5

27.2 Type Reference

27.2.1 InstanceId

int

27.2.2 IpStatus

struct

id	InstanceId
tico	optional bool
sfp	optional sfpStatus.SfpStatus
ipStreamStatuses	optional list of ipStreamStatus.IpStreamStatus
alternateInputStatus	optional alternateInputStatus.AlternateInputStatus

28 ipStream (1.3)

28.1 Overview

```
# Changelog
## 1.3
- **Changed**
- `st2110Types` from 1.0 to 1.2.
```

28.2 Type Reference

28.2.1 AlternateInput

IP config for a service using alternative input.

struct

path1	IpStreamGroup Config for path one.
path2	IpStreamGroup Config for path two.
bufferSize	int Input buffer size in milliseconds.
seamlessType	SeamlessType Config for the different types of alternative input.

28.2.2 Connection

A redundant Ip Connection.

variant

direct	IpStreamGroup No redundancy.
alternateInput	AlternateInput Seamless or alternative input.

28.2.3 EssenceProperties

Essence-specific properties of a 2110 essence.

variant

audio	St2110AudioProperties Properties of an audio essence.
-------	---

28.2.4 IpStream

This struct stores the address and other configurable options of an input or output stream on an IP network. On 2110 products it represents an essence.

struct

label	string A descriptive label or name for this stream.
address	firewallTypes.SocketAddress The IP address of this stream.
igmpv3Source	optional firewallTypes.IpAddress IP address of the IGMP source used for source filtering.
payloadId	int Id of the essence.
st2110Info	optional St2110Info Information which is only available for 2110 applications.
id	UUID Unique id within an IpStreamGroup. Redundant streams (in different groups) have the same id.

28.2.5 IpStreamGroup

A group of IpStreams that share the same physical port.

struct

bufferSize	optional int Size of buffer used when synchronizing essences.
source	list of IpStream The list of IpStreams in the group. Shall contain a single entry for 2022, but may contain multiple for 2110.
physicalPortId	physicalIpPort.PortName The id of the physical port on which the IpStreams are sent/received.

28.2.6 NearSeamless

Config for near seamless mode.

struct

switchingMode	SwitchingMode The mode of switching.
timeoutMainToSpare	int Timeout before switching to spare in milliseconds.
timeoutSpareToMain	int Timeout before switching to main in milliseconds.

28.2.7 SeamlessType

Config for seamless.

variant

smpte2022_7	SeamlessType.smpte2022_7 Config for SMPTE 2022-7 seamless.
nearSeamless	NearSeamless Config for near seamless mode.

28.2.8 SeamlessType.smpte2022_7

empty **struct**

28.2.9 St2110AudioProperties

Properties specific to a 2110 audio essence. Either -30 or -31.

struct

totalChannels	int The number of audio channels in the essence.
packetTime	st2110Types.St2110AudioPacketTime Audio packet time.

28.2.10 St2110Info

Information regarding any kind of 2110 essence.

struct

standard	st2110Types.St2110Standard The standard which the essence content follows.
properties	EssenceProperties Properties specific to a type of essence.

28.2.11 SwitchingMode

The switching modes that can be used with alternative input.

enum

PREFER_PATH1
PREFER_PATH2
FLOATING
FORCE_PATH1
FORCE_PATH2

29 ipStreamStatus (1.1)

29.1 Type Reference

29.1.1 IpStreamStatus

Represents the status of an IpStream.

struct

id	UUID The id of the IpStream this status represents.
bitrate	float The bitrate of the IpStream.
networkProtocol	NetworkProtocol Protocol used for seamless input.
sequenceErrors	int Number of gaps in RTP sequence numbers.
rtpSsrc	int Synchronization Source that uniquely identifies the source of a stream. Ssrc's within an RTP session will be unique.
minDelay	int Minimum network latency. [ms]
maxDelay	int Maximum network latency. [ms]

29.1.2 NetworkProtocol

Possible protocols of a seamless input.

enum

RTP	
SequencedUDP	
UDP	
Unknown	

30 ipStreamUtils (0.0)

31 linkModes (1.3)

31.1 Type Reference

31.1.1 IpFecMode

enum

NO_FEC

RS_FEC

FC_FEC

31.1.2 IpLinkMode

enum

IP_10G

IP_25G_COPPER

IP_25G_OPTICAL

IP_40G

IP_4x10G

31.1.3 IpLinkSpeed

enum

IP_10G

IP_25G

IP_40G

IP_4x10G

NO_LINK

32 lldp (1.1)

32.1 Overview

```
# Changelog
```

```
## 1.1
```

```
- **Changed**
```

```
- Extracted the structure inside of the `LldpNeighbor` object definition into a new type cal
```

32.2 Type Reference

32.2.1 LldpMedInventory

LLDP-MED Inventory. Appear specific implementation in parentheses.

struct

hwRev	string Hardware Revision
swRev	string Software Revision (Software Version)
fwRev	string Firmware Revision (FPGA Version)
serial	string Serial Number
manufacturer	string Manufacturer ("Appear")
model	string Model ("X10" or "X20")
assetId	string Asset ID (Hardware ID)

32.2.2 LldpMode

Choose either private or public configuration of LLDP frames.

enum

PUBLIC	Only mandatory TLVs and port detail TLVs.
PRIVATE	LLDP-MED inventory in addition to the PUBLIC TLVs.

32.2.3 LldpNeighbor

LldpNeighborStruct

32.2.4 LldpNeighborStruct

The optional variables are only present if `GetLldpNeighborRequest.details` is set to true in the rpc request. The following descriptions are the Appear specific implementations of each LLDP TLV. If neighbors from other manufacturers are detected, TLV values might differ. The `LldpMode` is specified in

parentheses for variables that have different implementations based on the LldpMode set in the neighbor.

struct

chassisId	string MAC address of CTRL on MMI 1, fallback to CTRL on MMI 2 (PRIVATE) or "N/A" (PUBLIC).
mgmtIp	list of string IPv4 address of CTRL on MMI 1 and/or CTRL on MMI 2. Also IPv6 if enabled on the neighbor.
sysDescr	optional string Desktop Heading (PRIVATE) or "Unknown" (PUBLIC)
sysName	optional string Hostname (PRIVATE) or Interface.Port (PUBLIC)
portId	string MAC address of local port
portDescr	optional string Slot, Interface: Port-label
ttl	optional string Time-To-Live, how long an LLDP packet received from that neighbor should be considered valid.
lldpMedInventory	optional LldpMedInventory LLDP-MED Inventory

33 lockStatus (1.3)

33.1 Overview

Changelog:

1.3

- **Changed**
 - poolCtrlAllocs from 1.0 to 1.1
- **Added**
 - flow from 1.0 to 1.1

33.2 Type Reference

33.2.1 GenlockStatus

struct

locked	bool
offset	int
validMode	bool
localReference	ReferenceLockStatus
remoteReference	ReferenceLockStatus

33.2.2 LockStatus

variant

genlockStatus	GenlockStatus
videoAlignmentStatus	VideoAlignmentStatus

33.2.3 ReferenceLockStatus

enum

UNLOCKED
LOCKED
HOLDOVER

33.2.4 VideoAlignmentStatus

struct

locked	bool
offset	int
group	int
alignmentMaster	poolCtrlAllocs.SlotFlow
outOfAlignment	bool
flow	int

34 logInsertion (1.0)

34.1 Overview

```
# Changelog

## 1.0
- **Added**
  - First version of module
```

34.2 Type Reference

34.2.1 LogInsertion

Configure logo insertion.

struct

id

string

A unique identifier for the logo. This is the value returned by the `imageUpload.SetImage` RPC.

xOffset

int

X offset from the top left corner in pixels.

yOffset

int

Y offset from the top left corner in pixels.

35 movedService (1.0)

35.1 Overview

```
# Changelog

## 1.0
- **Added**
  - added MovedServiceStruct and Obj
  - added slot type
```

35.2 Type Reference

35.2.1 MovedService

MovedServiceStruct

35.2.2 MovedServiceStruct

Struct that stores where services are being moved during service redundancy switch. For example in N+M redundancy mode

struct

serviceId	UUID Id of the service which will be moved
fromSlot	Slot From which slot the service will be moved
toSlot	optional Slot To which slot the service will be moved

35.2.3 Slot

Slot used by the service in the MovedServiceStruct

int

36 multiService (2.10)

36.1 Overview

Changelog

2.10

- **Changed**
 - added list of vanc components to MultiService
 - coderService from 2.37 to 2.38

2.9

- **Changed**
 - coderService from 2.36 to 2.37

2.8

- **Changed**
 - tsSource from 1.3 to 1.4

2.7

- **Changed**
 - coderService from 2.34 to 2.35
 - tsSource from 1.2 to 1.3
 - ipService from 1.1 to 1.2

2.6

- **Added** -logoInsertion 1.0
- **Changed**
 - coderService from 2.33 to 2.34

2.5

- **Changed**
 - coderService from 2.32 to 2.33

2.4

- **Changed**
 - coderService from 2.31 to 2.32

2.3

- **Changed**
 - coderService from 2.30 to 2.31

2.2

- **Added**
 - Scte35Config 1.0
- **Changed**

- `idrOnScte35` bool is now moved from `MultiServiceOptions` to `Scte35Options` in `scte35Config 1.0`
- `scte35` contains `Scte35Options` instead of `idrOnmScte35` bool
- `coderService` from 2.29 to 2.30

2.1

- **Added**

- `IpSource` struct containing an `IpService` and a list of slots in use

- **Changed**

- `coderService` from 2.28 to 2.29
- `ipService` from 1.0 to 1.1
- added `colorComponent` import and `colorProfile` to `MultiService`
- replaced `IpService` as `ip` variant in `MsSource` with newly added `IpSource`

2.0

- **Changed**

- `coderService` from 2.27 to 2.28
- added `ipService` as a possible multiservice source
- moved `audioOptions` to `MultiServiceOptions`
- replaced `audioOptions` on `ServiceOptions` with `audioSubscriptions` to refer to shared `audioOptions`
- these changes are not backwards compatible

1.22

- **Added**

- `latencyAdjustment` field added to `AudioOptions` struct for forwarding audio earlier than video.

- **Changed**

- `coderService` from 2.26 to 2.27

1.21

- **Changed**

- `coderService` from 2.25 to 2.26

36.2 Command Reference

36.2.1 SetMultiServices

- message **SetMultiServices.Request**
- message **SetMultiServices.Response**

36.2.2 GetMultiServices

- message **GetMultiServices.Request**
- message **GetMultiServices.Response**

36.2.3 DeleteMultiServices

- message **DeleteMultiServices.Request**
- message **DeleteMultiServices.Response**

36.3 Type Reference

36.3.1 AudioOptions

Audio options for a given Transcoder or ABR encoder audio track.

struct

source	coderService.AudioSource Source configuration for a track.
numAuPerPes	optional int Number of access units per PES, commonly referred to as PES alignment.
lipSyncAdjustment	optional int Lip sync adjustment
latencyAdjustment	optional int Adjust latency of audio relative to video.
levelAdjustment	optional int Level adjustment.
loudness	optional coderService.Loudness Loudness configuration.
destination	optional coderService.TsAudio Destination configuration for an ABR encoder.
passthrough	optional bool Flag based on the presence of an audio profile in the multi service profile for a given audio track; missing profiles indicates passthrough (true) here. ABR encoder only.

36.3.2 DeleteMultiServices.Request

struct

ids **list of UUID**

36.3.3 DeleteMultiServices.Response

empty **struct**

36.3.4 GetMultiServices.Request

empty **struct**

36.3.5 GetMultiServices.Response

struct

data	map from UUID to MultiServiceObj
info	GetMultiServices.Response.info

36.3.6 GetMultiServices.Response.info

struct

totalEntries	int
--------------	-----

36.3.7 IpSource

Configuration for IP sources.

struct

ipService	ipService.IpService Configuration for one or more IP video sources.
slot	list of int A list of slots utilizing the source.

36.3.8 MsSource

Multi Service Source. Configuration of the source of the Multi Service. Can be either ts, sdi or ip

variant

ts	tsSource.TsSource Configuration of the TS source. Transcoder only
sdi	SdiSource Configuration of the SDI source. ABR Encoder only
ip	IpSource Configuration of the IP source.

36.3.9 MultiService

Configuration for a Transcoder or ABR Encoder multi service.

struct

label	string A label for the service.
enabled	bool True if the service should be enabled.
source	MsSource Configuration of the source.
profile	MultiService.profile The profile that describes the Multi Service
options	MultiServiceOptions Configuration of components and IDR on SCTE35.
	optional testGenerator.SignalLoss

signalLoss	Configuration for the output when input signal is lost. ABR Encoder only
testPatternGenerator	optional testGenComponent.TestGenComponent Contains optional structure with link to a test generator profile. ABR Encoder only.
colorProfile	optional colorComponent.ColorComponent Contains optional structure with link to a color profile.
vancs	list of coderService.VancComponent A list of vanc components. SDI ABR Encoder only.

36.3.10 MultiService.profile

struct

id	UUID
----	------

36.3.11 MultiServiceObj

MultiService

36.3.12 MultiServiceOptions

Options for a given Transcoder or ABR encoder service.

struct

scte35	scte35Config.Scte35Config Configuration of SCTE 35.
audioOptions	map from int to AudioOptions Configuration of audio coding instances.
serviceOptions	map from int to ServiceOptions Configuration of component services.

36.3.13 SdiSource

Configuration for an ABR encoder.

struct

sdiConnection	sdiConnection.SdiConnection Configuration of the SDI connection of the source
slot	int Which slot the source comes from.

36.3.14 ServiceOptions

- Service options for a specific Transcoder or ABR encoder service.

•

struct

videoOptions

optional VideoOptions

Video options (unused).

•

audioSubscriptions

list of int

References to the audio tracks selected for this service. These identifiers must exist as keys in the map of AudioOptions.

•

passthroughComponents

optional bool

On transcoder only, setting this flag to true passes through all whitelisted transport stream components that are not transcoded by the service.

36.3.15 SetMultiServices.Request

struct

data

map from **UUID** to **MultiServiceObj**

36.3.16 SetMultiServices.Response

empty **struct**

36.3.17 VideoOptions

struct

logoInsertion

optional logoInsertion.LogoInsertion

Contains optional structure with configuration for logo insertion.

37 multiServiceProfile (2.3)

37.1 Overview

Changelog

2.3

- Changed
 - audioProfile from 2.4 to 2.5

2.2

- Changed
 - videoProfile from 2.8 to 2.9
 - expectedInput from 1.2 to 1.3

2.1

- Changed
 - audioProfile from 2.3 to 2.4

2.0

- Changed
 - moved audios to MultiServiceProfile
 - added audioSubscriptions to SubService to refer to shared audios
 - these changes are not backwards compatible

1.10

- Changed
- AudioTrack.profile field is changed from AudioProfileStruct to optional.
- expectedInput from 1.1 to 1.2

37.2 Command Reference

37.2.1 SetMultiServiceProfiles

- message **SetMultiServiceProfiles.Request**
- message **SetMultiServiceProfiles.Response**

37.2.2 GetMultiServiceProfiles

- message **GetMultiServiceProfiles.Request**
- message **GetMultiServiceProfiles.Response**

37.2.3 DeleteMultiServiceProfiles

- message **DeleteMultiServiceProfiles.Request**
- message **DeleteMultiServiceProfiles.Response**

37.3 Type Reference

37.3.1 AudioTrack

Audio Track configuration parameters.

@param uid An audio track's unique id. Values are not required to be contiguous.

@param profile A profile describing an audio encoding process. Empty optional signals passthrough

struct

uid	int
profile	optional audioProfile.AudioProfileStruct

37.3.2 DeleteMultiServiceProfiles.Request

struct

ids	list of UUID
-----	---------------------

37.3.3 DeleteMultiServiceProfiles.Response

empty **struct**

37.3.4 GetMultiServiceProfiles.Request

empty **struct**

37.3.5 GetMultiServiceProfiles.Response

struct

data	map from UUID to MultiServiceProfileObj
info	GetMultiServiceProfiles.Response.info

37.3.6 GetMultiServiceProfiles.Response.info

struct

totalEntries	int
--------------	-----

37.3.7 MultiServiceProfile

Multiservice configuration parameters.

@param profileName A text identifier for the profile.

@param profileMode Flag for broadcast or ABR profile modes.

@param input Describes the expected input format of the video signal.

@param gopSize Describes the distances in frames between two keyframes.

@param aspectRatio Describes the ratio of width to height of the video images.

@param frameRateDomain Describes the restricted set of frame rates these services should support

@param audios List of audio configurations for this profile. Each AudioTrack can be used in mult

@param subServices List of subservice definitions.

struct

profileName	string
profileMode	multiServiceProfileMode.ProfileMode
input	expectedInput.ExpectedInput
gopSize	int
aspectRatio	videoProfile.AspectRatio
frameRateDomain	rawVideo.FrameRate
audios	list of AudioTrack
subServices	list of SubService

37.3.8 MultiServiceProfileObj

MultiServiceProfile

37.3.9 SetMultiServiceProfiles.Request

struct

data map from UUID to MultiServiceProfileObj

37.3.10 SetMultiServiceProfiles.Response

empty struct

37.3.11 SubService

Subservice configuration parameters.
 @param index Subservice unique id. Values are not required to be contiguous.
 @param video Video parameters.
 @param audioSubscriptions List of AudioTrack uid's to include in this subservice. These must be

struct

index	int
video	optional VideoTrack
audioSubscriptions	list of int

37.3.12 VideoTrack

Video Track configuration parameters.
 @param profile Describes a video coding process.

struct

profile videoProfile.VideoProfileStruct

38 multiServiceProfileMode (1.0)

38.1 Type Reference

38.1.1 ProfileMode

enum

ABR

BROADCAST

39 nmosConfig (1.1)

39.1 Overview

```
# Changelog

## 1.1
- **Added**
  - Added new type `ConnectionType`
  - Added member 'connectionType' to NmosConfig
```

39.2 Type Reference

39.2.1 ConnectionType

Enum to control whether the NMOS traffic is in-band (through line card dataports) or out-of-band (through the MMI).

enum

```
IN_BAND
OUT_OF_BAND
```

39.2.2 NmosConfig

Configuration parameters for setting up NMOS.

struct

connectionType	ConnectionType Configuration to control whether traffic should be in-band or out-of-band
registry	optional nmosRegistryConfig.NmosRegistryConfigStruct Configuration parameters for connecting to the NMOS registry
labelTemplates	optional nmosLabelConfig.NmosLabelConfig Custom NMOS labels configuration. This feature is enabled when the optional is set

40 nmosLabelConfig (1.0)

40.1 Overview

Changelog

1.0

NMOS labelling

NMOS labelling API allows operator to define templates based on auto text options. This templates can be defined for node, device, source, flow, sender and receiver.

Version 1.0 allows six auto text options.

What are the auto text options?

- `${SLOT}` – The slot number of the card in use.
- `${DEV_ID}` – The Device number. **Not supported for Node templates.**
- `${ESS_GR}` – The label of the group where the essences belong to. **Not supported for Node templates.**
- `${ESS}` – The label of the Essence in question. **Not supported for Node and Device templates.**
- `${ESS_T}` – The Essence type: Video, Audio, Ancillary. This auto text option is substituted by what is defined in `EssenceTypeStrings` for each of the essences type. **Not supported for Node and Device templates.**
- `${ESS_T_INDEX}` – The number of the essence belonging to one of the essence type – video, audio, ancillary. **Not supported for Node and Device templates.**

How to use the NMOS Labelling.

- NMOS templates are set in the fields part of `ResourceLabelTemplates`. Unused templates should be set with an empty string.
- Each field in `ResourceLabelTemplates` can have as many auto text options as the operator wants, as long as they are supported for the respective template.
- `EssenceTypeStrings` strings will substitute the auto text option `${ESS_T}` with the defined string for the respective Essence type.

Example of strings to use for the fields in `ResourceLabelTemplates` and `EssenceTypeStrings`

- `nodeTemplate` – ECx210 Decoder ST2110 Sender Slot `${SLOT}`
- `deviceTemplate` – ECx210 Decoder Sender Slot `${SLOT}` ID `${DEV_ID}`
- `sourceTemplate` – ECx210 Slot `${SLOT}` ID `${DEV_ID}`, `${ESS_T}` Source: `${ESS}`, Group: `${ESS_GR}`, Essence Number: `${ESS_T_INDEX}`
- `flowTemplate` – ECx210 Slot `${SLOT}` ID `${DEV_ID}`, `${ESS_T}` Flow: `${ESS}`, Group: `${ESS_GR}`, Essence Number: `${ESS_T_INDEX}`
- `senderTemplate` – ECx210 Slot `${SLOT}` ID `${DEV_ID}`, `${ESS_T}` Sender: `${ESS}`, Group: `${ESS_GR}`, Essence Number: `${ESS_T_INDEX}`
- `receiverTemplate` – ECx210 Slot `${SLOT}` ID `${DEV_ID}`, `${ESS_T}` Receiver: `${ESS}`, Group: `${ESS_GR}`, Essence Number: `${ESS_T_INDEX}`
- `video` – Video
- `audio` – Audio
- `ancillary` – Anc

40.2 Type Reference

40.2.1 EssenceTypeStrings

Labels for each essence type to use when generating custom NMOS labels.

struct

video	string Video essence label
audio	string Audio essence label
ancillary	string Ancillary essence label

40.2.2 NmosLabelConfig

Custom NMOS labels configuration.

struct

resourceTemplates	ResourceLabelTemplates Templates for each NMOS resource
essenceStrings	EssenceTypeStrings Labels to use for each essence type

40.2.3 ResourceLabelTemplates

NMOS label templates for each NMOS resource.

struct

nodeTemplate	string Template string for the node label
deviceTemplate	string Template string for the device label
sourceTemplate	string Template string for the source label
flowTemplate	string Template string for the flow label
senderTemplate	string Template string for the sender label
receiverTemplate	string Template string for the receiver label

4.1 nmosRegistryConfig (1.0)

4.1.1 Type Reference

4.1.1.1 DnsConfig

Method to use for discovery of registry server

variant

mDns	DnsConfig.mDns Use mDNS to reach the registry server
dnsSd	DnsSdConfig Configuration parameters to use DNS-SD for discovery of registry service

4.1.1.2 DnsConfig.mDns

empty **struct**

4.1.1.3 DnsSdConfig

Configuration parameters for DNS-SD

struct

dnsServer	firewallTypes.IpAddress IPv4 address of DNS-SD server
search	string Search domain name

4.1.1.4 NmosRegistryConfigStruct

Configuration parameters for reaching the NMOS registry

struct

registryConfig	RegistryManualConfig Configuration parameters to use in case of failure in mDNS or DNS-SD (secondary/fallback method)
dnsConfig	DnsConfig Configuration parameters for mDNS or DNS-SD (primary method)

4.1.1.5 RegistryManualConfig

Registry information to be used in case registry server can not be reached by mDNS or DNS-SD.

struct

fallbackEnable	bool Enable use of given information as a fallback
	firewallTypes.IpAddress

registryIpAddr	IPv4 address of NMOS registry
registryPort	int Port number of NMOS registry
registryVersion	commonTypes.NmosVersionNumber Version of registry API

42 nmosRegistryStatus (1.0)

42.1 Type Reference

42.1.1 NmosRegistryStatus

NMOS registry status

struct

ipAddr

optional **firewallTypes.SocketAddress**

Currently used IP address and port number of the registry server. This value may have been obtained from DNS-SD, mDNS, or a manual IP entry.

registered

bool

True if there is an active connection to the registry server.

43 nmosStatusTypes (1.1)

43.1 Type Reference

43.1.1 NmosStatus

NmosStatus Status of the NMOS service

struct

nmosRegistryStatus	optional nmosRegistryStatus.NmosRegistryStatus Status of the NMOS registry connection. Empty if no registry is configured.
nodeApi	map from physicalIpPort.PortName to string Map from port to the endpoint for NMOS Node API. Ex: PortName.D1 -> "10.20.10.34:8001", PortName.D2_2 -> "10.20.11.34:8001"

44 physicalpPort (1.0)

44.1 Type Reference

44.1.1 PortName

enum

CTRL

D1

D2

D3

D4

D1_1

D1_2

D1_3

D1_4

D2_1

D2_2

D2_3

D2_4

D3_1

D3_2

D3_3

D3_4

D4_1

D4_2

D4_3

D4_4

45 poisServerStatus (1.0)

45.1 Overview

```
# Changelog

## Version 1.0
- **Added**
  - `PoisServerStatus`
  - `GetPoisServerStatus`
```

45.2 Command Reference

45.2.1 GetPoisServerStatus

- message **GetPoisServerStatus.Request**
- message **GetPoisServerStatus.Response**

45.3 Type Reference

45.3.1 GetPoisServerStatus.Request

empty **struct**

45.3.2 GetPoisServerStatus.Response

struct

data **PoisServerStatus**

45.3.3 PoisServerStatus

Status of POIS server

@param connectionStatus Status of the connection to the server

struct

connectionStatus **esamStatus.PoisConnectionStatus**

46 poolConfig (1.1)

46.1 Overview

Changelog

1.1

- Changed
 - included esamConfig in PoolConfig struct

46.2 Command Reference

46.2.1 GetPoolConfig

- message **GetPoolConfig.Request**
- message **GetPoolConfig.Response**

46.2.2 SetPoolConfig

- message **SetPoolConfig.Request**
- message **SetPoolConfig.Response**

46.3 Type Reference

46.3.1 GetPoolConfig.Request

empty **struct**

46.3.2 GetPoolConfig.Response

struct

data **PoolConfigObj**

46.3.3 PoolConfig

struct

spareModules	optional int
esamConfig	optional esamConfig.EsamConfig

46.3.4 PoolConfigObj

PoolConfig

46.3.5 SetPoolConfig.Request

struct

data

PoolConfigObj

46.3.6 SetPoolConfig.Response

empty struct

47 poolCtrlAllocs (1.1)

47.1 Overview

Changelog:

1.1

- **Added**
 - Constant base UUID used to generate deterministic UUIDs for video alignment groups.
- **Changed**
 - generalization of the video alignment in order to work for both SDI and IP: VideoAlignGroupMaster, VideoAlignGroupMasterObj and FlowStatus should replace VideoAlignGroupMasters, Allocations and Distribution
 - SlotPort with member port changed to SlotFlow with member flow

47.2 Type Reference

47.2.1 SlotFlow

struct

slot	int
flow	int

47.2.2 VideoAlignGroupMaster

struct

groupIndex	int
master	UUID
flow	SlotFlow

47.2.3 VideoAlignGroupMasterObj

VideoAlignGroupMaster

48 psiStatus (1.6)

48.1 Overview

```
# Changelog

## 1.6
- **Changed**
  - `psiTypes` from 1.5 to 1.6

## 1.5
- **Changed**
  - `psiTypes` from 1.4 to 1.5
```

48.2 Type Reference

48.2.1 PsiStatus

struct

components	list of psiTypes.Component
name	optional string

49 psiTypes (1.6)

49.1 Overview

```
# Changelog

## 1.6
- **Changed**
  - Component.bitrate changed from int to bigint. Necessary for UHD

## 1.5
- **Changed**
  - `component` from 1.3 to 1.5
```

49.2 Type Reference

49.2.1 Component

struct

flowId	int
pid	Pid
lang	Lang
componentType	component.ComponentType
streamType	int
isPcr	bool
cas_ids	list of int
ccError	int
bitrate	bigint

49.2.2 Components

list of **Component**

49.2.3 Lang

string

49.2.4 Pid

int

50 rawAudio (1.1)

50.1 Type Reference

50.1.1 ChannelMode

enum

MONO

DUALMONO

STEREO

SURROUND_31

SURROUND_51

SURROUND_71

50.1.2 Hack

struct

profile

int

51 rawVideo (1.4)

51.1 Type Reference

51.1.1 ChromaSampling

enum

CS_422

CS_420

51.1.2 FrameRate

enum

FPS_25

FPS_29_97

FPS_30

FPS_50

FPS_59_94

FPS_60

51.1.3 Hack

struct

profile int

51.1.4 HorizontalResolution

enum

H_3840

H_2560

H_1920

H_1440

H_1280

H_1024

H_960

H_854

H_848

H_768

H_720

H_704

H_640

H_544

H_528

H_512

H_480

H_400

H_384	
H_320	
H_256	
H_192	

51.1.5 VerticalResolution

enum

V_2160	
V_1440	
V_1080	
V_720	
V_640	
V_576	
V_540	
V_480	
V_432	
V_396	
V_384	
V_360	
V_288	
V_270	
V_224	
V_216	
V_192	
V_180	
V_144	

52 redundancyGroup (1.0)

52.1 Overview

```
# Changelog

## 1.0
- **Added**
  - added RedundancyMode
  - added RedundancyType
  - added RedundancyCard
  - added RedundancyGroupStruct
```

52.2 Command Reference

52.2.1 GetRedundancyGroups

- message **GetRedundancyGroups.Request**
- message **GetRedundancyGroups.Response**
- message **GetRedundancyGroups.Failure**

52.2.2 SetRedundancyGroups

- message **SetRedundancyGroups.Request**
- message **SetRedundancyGroups.Response**
- message **SetRedundancyGroups.Failure**

52.2.3 DeleteRedundancyGroups

- message **DeleteRedundancyGroups.Request**
- message **DeleteRedundancyGroups.Response**
- message **DeleteRedundancyGroups.Failure**

52.3 Type Reference

52.3.1 DeleteRedundancyGroups.Failure

empty **struct**

52.3.2 DeleteRedundancyGroups.Request

struct

ids **list** of **UUID**

52.3.3 DeleteRedundancyGroups.Response

empty **struct**

52.3.4 GetRedundancyGroups.Failure

empty **struct**

52.3.5 GetRedundancyGroups.Request

empty **struct**

52.3.6 GetRedundancyGroups.Response

struct

data **map** from **UUID** to **RedundancyGroup**

52.3.7 RedundancyCard

N+M redundancy configuration for a single card

struct

slot	int The slot the card resides in
redundancyType	RedundancyType Specifies whether the card is a main or backup card
disabled	bool True if the card has been disabled

52.3.8 RedundancyGroup

RedundancyGroupStruct

52.3.9 RedundancyGroupStruct

N+M redundancy group configuration

struct

label	string A label for this group
router	optional UUID The SDI router connected to the cards in this group
mode	RedundancyMode The redundancy mode used for this group
cards	list of RedundancyCard List of cards that are part of this group

52.3.10 RedundancyMode

The supported N+M redundancy modes

enum

FLOATING	
REVERTING	

52.3.11 RedundancyType

The roles a card can have in a redundancy group

enum

MAIN	
BACKUP	

52.3.12 SetRedundancyGroups.Failure

empty struct

52.3.13 SetRedundancyGroups.Request

struct

data map from UUID to RedundancyGroup

52.3.14 SetRedundancyGroups.Response

empty struct

53 redundancyGroupStatus (1.2)

53.1 Overview

```
# Changelog

## 1.2
- **Added**
  - added redundancyStateToString

## 1.1
- **Added**
  - added ProtoRedundancyGroupStatus
```

53.2 Command Reference

53.2.1 GetRedundancyGroupStatus

- message **GetRedundancyGroupStatus.Request**
- message **GetRedundancyGroupStatus.Response**

53.3 Type Reference

53.3.1 GetRedundancyGroupStatus.Request

empty **struct**

53.3.2 GetRedundancyGroupStatus.Response

struct

data **map** from **UUID** to **RedundancyGroupStatus**

53.3.3 RedundancyCardStatus

N+M redundancy status for a single card

struct

state	RedundancyState The current state of the card
activeConfig	optional Slot The active configuration of the card. A slot means that the services on that slot was moved to this card.

53.3.4 RedundancyGroupStatus

N+M redundancy status for a redundancy group

struct

map from **Slot** to **RedundancyCardStatus**

status

Redundancy status for each card in this group

53.3.5 RedundancyState

The states a single card can be in.

enum

UP

DOWN

DISABLED

53.3.6 Slot

int

54 scte35Config (1.0)

54.1 Type Reference

54.1.1 Scte35Config

Config for a Scte35 configuration.

struct

idrOnScte35

bool

Enables IDR insertion on SCTE 35 splice points.

esamConfig

optional **esamConfig.EsamServiceConfig**

ESAM configuration. A set optional indicates that ESAM is enabled.

55 scte35LogApi (1.0)

55.1 Command Reference

55.1.1 GetScte35History

- message **GetScte35History.Request**
- message **GetScte35History.Response**

55.2 Type Reference

55.2.1 GetScte35History.Request

struct

serviceId	UUID
slot	int

55.2.2 GetScte35History.Response

struct

scte35History	string
---------------	--------

56 sdiConnection (1.0)

56.1 Overview

New in v1.0:

- These types were factored out from coderService

56.2 Type Reference

56.2.1 SdiConnection

struct

sdiType	sdiType.SdiType
connectors	SdiConnectors

56.2.2 SdiConnectors

enum

SDI_1A	
SDI_1B	
SDI_1C	
SDI_1D	
SDI_2A	
SDI_2B	
SDI_2C	
SDI_2D	
SDI_1ABCD	
SDI_2ABCD	
SDI_1AC	
SDI_1BD	
SDI_2AC	
SDI_2BD	
UNDEFINED	

57 sdiRouterConfig (1.1)

57.1 Overview

```
# Changelog
## 1.1
- **Added**
  - added brand SWP08
  - added port to SdiRouterConfig

## 1.0
- **Added**
  - added SdiRouterStruct
  - added SdiRouterBrand
  - added CardConnector
```

57.2 Command Reference

57.2.1 GetSdiRouters

- message **GetSdiRouters.Request**
- message **GetSdiRouters.Response**
- message **GetSdiRouters.Failure**

57.2.2 SetSdiRouters

- message **SetSdiRouters.Request**
- message **SetSdiRouters.Response**
- message **SetSdiRouters.Failure**

57.2.3 DeleteSdiRouters

- message **DeleteSdiRouters.Request**
- message **DeleteSdiRouters.Response**
- message **DeleteSdiRouters.Failure**

57.3 Type Reference

57.3.1 CardConnector

Holds slot and connector which is enough to specify a distinct input port in a chassis.

struct

slot	int
connector	sdiConnection.SdiConnectors

57.3.2 DeleteSdiRouters.Failure

empty **struct**

57.3.3 DeleteSdiRouters.Request

struct

ids **list** of **UUID**

57.3.4 DeleteSdiRouters.Response

empty **struct**

57.3.5 GetSdiRouters.Failure

empty **struct**

57.3.6 GetSdiRouters.Request

empty **struct**

57.3.7 GetSdiRouters.Response

struct

data **map** from **UUID** to **SdiRouter**

57.3.8 Label

string

57.3.9 Port

int

57.3.10 SdiRouter

SdiRouterStruct

57.3.11 SdiRouterBrand

The different SDI router brands we support

enum

BLACKMAGIC

NEVION

SWP08

57.3.12 SdiRouterStruct

SDI router configuration

struct

label	string A label for the SDI router
ip	firewallTypes.IpAddress The IP address of the router
port	optional int The port of the router
brand	SdiRouterBrand The router brand
numInputs	int The number of SDI inputs the router has
numOutputs	int The number of SDI outputs the router has
inputs	map from Port to Label Label for each input port
outputs	map from Port to Label Label for each output port
outputMapping	map from Port to CardConnector Maps router output port to coder slot and connector

57.3.13 SetSdiRouters.Failureempty **struct****57.3.14 SetSdiRouters.Request****struct**data **map** from **UUID** to **SdiRouter****57.3.15 SetSdiRouters.Response**empty **struct**

58 sdiRouterConnection (1.0)

58.1 Overview

```
# Changelog

## 1.0
- **Added**
  - added SdiRouterConnectionStruct
```

58.2 Command Reference

58.2.1 SetSdiRouterConnections

- message **SetSdiRouterConnections.Request**
- message **SetSdiRouterConnections.Response**
- message **SetSdiRouterConnections.Failure**

58.3 Type Reference

58.3.1 InputPort

int

58.3.2 OutputPort

int

58.3.3 SdiRouterConnectionStruct

Keeps track of the internal routing in a SDI router

struct

routerId	UUID The ID of the SDI router
routing	map from OutputPort to InputPort Map of which input is available on each output

58.3.4 SetSdiRouterConnections.Failure

empty **struct**

58.3.5 SetSdiRouterConnections.Request

struct

data	map from UUID to SdiRouterConnectionStruct
------	---

58.3.6 SetSdiRouterConnections.Response

empty **struct**

59 sdiRouterStatus (1.0)

59.1 Overview

```
# Changelog

## 1.0
- **Added**
  - added SdiRouterConfigStatus
  - added GetSdiRouterConfigStatuses
```

59.2 Command Reference

59.2.1 GetSdiRouterStatuses

- message **GetSdiRouterStatuses.Request**
- message **GetSdiRouterStatuses.Response**

59.3 Type Reference

59.3.1 GetSdiRouterStatuses.Request

empty **struct**

59.3.2 GetSdiRouterStatuses.Response

struct

data **map** from **UUID** to **SdiRouterStatus**

59.3.3 InputPort

int

59.3.4 Label

string

59.3.5 OutputPort

int

59.3.6 SdiRouterStatus

Status for a connected SDI router

@param routing The currently configured routing for each output on the router
 @param inputLabels The labels for the inputs on the switch
 @param outputLabels The labels for the outputs on the switch

struct

routing	<code>map</code> from <code>OutputPort</code> to <code>InputPort</code>
inputLabels	<code>map</code> from <code>InputPort</code> to <code>Label</code>
outputLabels	<code>map</code> from <code>OutputPort</code> to <code>Label</code>
link	<code>bool</code>

60 sdiStatus (1.9)

60.1 Overview

```
# Changelog

## 1.9
- **Added**
  - SdiAudioStatus analysedMaxLevel and analysedCodec.

## 1.8
- **Added**
  - SDI_2160P25, SDI_2160P30, SDI_2160P30_1001 to the SdiVideoModes enum
```

60.2 Type Reference

60.2.1 AesChannelMode

enum

MONO
STEREO
MULTI
N_A
UNKNOWN

60.2.2 AesData

struct

mode	AesMode
channelMode	AesChannelMode
crcOk	bool

60.2.3 AesMode

enum

PCM
NON_PCM

60.2.4 InstanceId

int

60.2.5 QuadMode

enum

SDI_4X3G_SQUARE
SDI_4X3G_INTERL

60.2.6 QuadStatus

struct

lock	bool
lockLoss	int
skew	float

60.2.7 SdiAudioSampleRate

enum

SYNC_48_kHz
ASYNC_48_kHz
SYNC_44_1_kHz
ASYNC_44_1_kHz
SYNC_32_kHz
ASYNC_32_kHz
SYNC_FREE_RUN
ASYNC_FREE_RUN

60.2.8 SdiAudioStatus

SDI input audio status on an encoder.

struct

aesChannel	int	Embedded audio channels.
active	bool	Indicates if a channel is active meaning that there exists a valid signal on the input.
lossCnt	int	Number of lost packets during the last second.
sampleRate	SdiAudioSampleRate	Audio sample rate.
sampleCnt	int	Actual number of samples received during the last second.
errors	bool	True if any errors during the last second, false otherwise.
aesData	optional AesData	Embedded audio channel data.
analysedMaxLevel	optional float	Analysed input max level.
analysedCodec	optional codedAudio.AudioCodec	Analysed input codec.

60.2.9 SdiStatus

SDI input status on an encoder, and SDI output status on a decoder.

struct

id	InstanceId Used with quad SDI input. IDs 0–3 correspond to signals on 1A–D and 4–7 correspond to 2A–D.
videoLockInput	bool True if the encoder has locked onto a signal, false otherwise.
videoMode	SdiVideoModes Input video mode on encoder, and output video mode on the decoder.
edhErrorCount	int Number of EDH errors on the input signal.
audio	list of SdiAudioStatus List of audio statuses for each audio channel in the SDI signal.
quad	optional QuadStatus Quad status for this SDI instance. Only available with quad input.
quadMode	optional QuadMode Which mode was used to generate the quad signal. Only available with quad input.
transceiverStatus	optional transceiverStatus.TransceiverStatus Status from the transceiver. Only available on encoder.

60.2.10 SdiVideoModes**enum**

SDI_OFF
 SDI_480I
 SDI_480P
 SDI_576I
 SDI_576P
 SDI_720P24
 SDI_720P24_1001
 SDI_720P25
 SDI_720P30
 SDI_720P30_1001
 SDI_720P50
 SDI_720P60
 SDI_720P60_1001
 SDI_1080P24
 SDI_1080P24_1001
 SDI_1080P25
 SDI_1080P30
 SDI_1080P30_1001
 SDI_1080I50
 SDI_1080I60
 SDI_1080I60_1001
 SDI_1080P50
 SDI_1080P60
 SDI_1080P60_1001

SDI_2160P50

SDI_2160P60

SDI_2160P60_1001

SDI_2160P25

SDI_2160P30

SDI_2160P30_1001

ANY

61 sdiType (1.0)

61.1 Type Reference

61.1.1 SdiType

enum

SDI_3G

SDI_12G

SDI_4X3G_QUANT

SDI_4X3G_SQUARE

SDI_4X3G_INTERL

62 sdpUpload (1.1)

62.1 Overview

This file defines the external RPC interface for uploading SDP files in order to change an IP Pr

Changelog

1.1

- ****Changed****
- `profileId` added to `SdpData` struct for use with Profile Redundancy

62.2 Command Reference

62.2.1 UploadSdpData

- message **UploadSdpData.Request**
- message **UploadSdpData.Response**

62.3 Type Reference

62.3.1 SdpData

- SDP data for storing the SDP file uploaded through GUI.

•

struct

serviceId

UUID

The uuid of the service for which the SDP file is uploaded.

•

profileId

optional UUID

The uuid of the profile for which the SDP file is uploaded. If profileId is empty, the uploaded SDP should belong to the first profile (index 0).

•

slot

int

The slot on which this SDP data will be used.

•

sdpInfo

map from UUID to string

The SDP file stored with a unique uuid.

62.3.2 UploadSdpData.Request

SdpData

62.3.3 UploadSdpData.Response

empty struct

63 serviceStatus (2.39)

63.1 Overview

Changelog:

2.39

- **Changed**
 - vancStatus from 1.10 to 1.11
 - audioStatus from 2.13 to 2.14
 - st2110Status from 1.3 to 1.4

2.38

- **Changed**
 - vancStatus from 1.9 to 1.10

2.37

- **Changed**
 - videoStatus from 2.11 to 2.12

2.36

- **Changed**
 - vancStatus from 1.8 to 1.9

2.35

- **Changed**
 - st2110Status from 1.2 to 1.3
 - audioStatus from 2.12 to 2.13
 - vancStatus from 1.7 to 1.8

2.34

- **Changed**
 - audioStatus from 2.11 to 2.12
 - sdiStatus from 1.8 to 1.9

2.33

- **Changed**
 - vancStatus from 1.6 to 1.7

2.32

- **Changed**
 - vancStatus from 1.5 to 1.6

2.31

- **Changed**
 - lockStatus from 1.2 to 1.3

2.30

- **Changed**
 - videoStatus from 2.10 to 2.11
 - psiStatus from 1.5 to 1.6
 - inputRedundancyStatus from 1.2 to 1.3

2.29

- **Added**
 - New struct St2110ProfileStatus added, replacing st2110 list in ServiceStatus

2.28

- **Changed**
 - sdiStatus from 1.7 to 1.8

2.27

- **Changed**
 - audioStatus from 2.10 to 2.11
 - psiStatus from 1.4 to 1.5
 - inputRedundancyStatus from 1.1 to 1.2

2.26

- **Changed**
 - ipStatus from 1.6 to 1.7

63.2 Command Reference

63.2.1 GetServiceStatus

- message **GetServiceStatus.Request**
- message **GetServiceStatus.Response**

63.3 Type Reference

63.3.1 GetServiceStatus.Request

ServiceStatusRequest

63.3.2 GetServiceStatus.Response

ServiceStatusResponse

63.3.3 ServiceStatus

Status for a specific service

struct

video	optional videoStatus.VideoStatus Status for resulting encoded or decoded video.
audio	optional audioStatus.AudioStatus Audio status
sdi	map from int to sdiStatus.SdiStatus SDI input status per channel (also used for IP-2022-Encoder)
st2110	optional st2110ProfileStatus ST-2110 specific status; list of essences and uuid of the active ip profile
lock	optional lockStatus.LockStatus Lock status (genlock etc.)
vanc	optional vancStatus.VancStatus Status for the supported VANC protocols (teletext etc.)
ip	map from int to ipStatus.IpStatus IP status
components	optional psiStatus.PsiStatus Status for component passthrough (transcoder)
latency	optional int Total latency of the service in ms (encoder only)
tsInputRedundancy	optional inputRedundancyStatus.TsInputRedundancyStatus

63.3.4 ServiceStatusQuery

struct

id	UUID
slot	int

63.3.5 ServiceStatusRequest

struct

query	list of ServiceStatusQuery
-------	--

63.3.6 ServiceStatusResponse

struct

data	map from UUID to ServiceStatus
------	---

63.3.7 St2110ProfileStatus

struct

st2110Essences	list of st2110Status.St2110Status
profileId	UUID

64 sfpConstants (1.4)

64.1 Overview

Changelog

1.4

- **Changed**
- Changed "SFP or SFP+" to "SFP/SFP+/SFP28" The corresponding identifier refers to the given 3 SFP types as per SNIA SFF-8024 specification.

64.2 Type Reference

64.2.1 Identifier

Physical Device Identifier Values. (SFF 8472 Rev12.3 spec Table 5-1)

enum

UNKNOWN	Unknown or unspecified
GBIC	GBIC
SOLDERED	Module soldered to motherboard (ex. SFF)
SFP_SFPP	SFP, SFP+ or SFP28
QSFP	QSFP+
QSFP28	QSFP28
NOT_USED	Not used by SFF 8472 Rev12.2.1. These values are maintained in the Transceiver Management section of SFF-8024.
VENDOR	Vendor specific

64.2.2 Transceiver

Transceiver Compliance Codes. (SFF 8472 Rev12.3 spec Table 5-3)

enum

ER_10G	10G Base-ER
LRM_10G	10G Base-LRM
LR_10G	10G Base-LR
SR_10G	10G Base-SR
ACTIVE_CABLE_40G	40G Active cable
LR4_40G	40G Base-LR4
SR4_40G	40G Base-SR4
CR4_40G	40G Base-CR4
PX	BASE-PX
BX10	BASE-BX10
FX_100	100BASE-FX
LX_100	100BASE-LX/LX10
T_1000	1000BASE-T
CX_1000	1000BASE-CX
LX_1000	1000BASE-LX
SX_1000	1000BASE-SX

ELECTRICAL_INTER_ENCLOSURE	Electrical inter enclosure
LONGWAVE_LASER_LL	Longwave laser LL
MEDIUM	Medium distance
LONG_DISTANCE	Long distance
INTERMEDIATE_DISTANCE	Intermediate distance
SHORT_DISTANCE	Short distance
VERY_LONG_DISTANCE	Very long distance
ACTIVE_CABLE	Active Cable
PASSIVE_CABLE	Passive Cable
LONGWAVE_LASER_LC	Longwave laser LC
SHORTWAVE_LASER_w_OFC	Shortwave laser w OFC
SHORTWAVE_LASER_wo_OFC	Shortwave laser w/o OFC
ELECTRICAL_INTRA_ENCLOSURE	Electrical intra enclosure
SINGLE_MODE	Singel mode
MULTIMODE_50um	Multimode 50um
MULTIMODE_50m	Multimode 50m
MULTIMODE_625m	Multimode 62.6m
VIDEO_COAX	Video coax
MINIATURE_COAX	Miniature coax
SHIELDED_TWISTED_PAIR	Shielded twisted pair
TWIN_AXIAL_PAIR	Twin axial pair
M_BYTES_SEC_100	100M bytes per sec
M_BYTES_SEC_200	200M bytes per sec
M_BYTES_SEC_400	400M bytes per sec
M_BYTES_SEC_1600	1600M bytes per sec
M_BYTES_SEC_800	800M bytes per sec
M_BYTES_SEC_1200	1200M bytes per sec
	Extended Specification Compliance. (SFF-8024 Rev 4.8 spec Table 4-4)
AOC_100G_AOC_25GAUI_C2M_BER_5_	100G AOC (Active Optical Cable) or 25GAUI C2M AOC. Providing a worst BER of $5 * 10^{-5}$
SR4_100G_BASE_SR_25G_BASE	100GBASE-SR4 or 25GBASE-SR
LR4_100G_BASE_LR_25G_BASE	100GBASE-LR4 or 25GBASE-LR
ER4_100G_BASE_ER_25G_BASE	100GBASE-ER4 or 25GBASE-ER
SR10_100G_BASE	100GBASE-SR10
CWDM4_100G	100G CWDM4
PSM4_PARALLEL_SMF_100G	100G PSM4 Parallel SMF
ACC_100G_ACC_25GAUI_C2M_BER_5_	100G ACC (Active Copper Cable) or 25GAUI C2M ACC. Providing a worst BER of $5 * 10^{-5}$
CR4_100G_BASE_25G_BASE_CR_CA_2	100GBASE-CR4 or 25GBASE-CR CA-L
CR_25G_BASE_CA_25G_S_50G_BASE_	25GBASE-CR CA-S
CR_25G_BASE_CA_25G_N_50G_BASE_	25GBASE-CR CA-N
ER4_40G_BASE	40GBASE-ER4
SR_4_X_10G_BASE	4 x 10GBASE-SR
PSM4_PARALLEL_SMF_40G	40G PSM4 Parallel SMF
G959_1_PROFILE_P1I1_2D1	G959.1 profile P1I1-2D1 (10709 MBd, 2km, 1310nm SM)
G959_1_PROFILE_P1S1_2D2	G959.1 profile P1S1-2D2 (10709 MBd, 40km, 1550nm SM)
G959_1_PROFILE_P1L1_2D2	G959.1 profile P1L1-2D2 (10709 MBd, 80km, 1550nm SM)
T_10G_BASE	10GBASE-T with SFI electrical interface

CLR4_100G	100G CLR4
AOC_100G_AOC_25GAUI_C2M_BER_10	100G AOC or 25GAUI C2M AOC. Providing a worst BER of 10^{-12} or below
ACC_100G_ACC_25GAUI_C2M_BER_10	100G ACC or 25GAUI C2M ACC. Providing a worst BER of 10^{-12} or below
DWDM2_100GE	100GE-DWDM2 (DWDM transceiver using 2 wavelengths on a 1550nm DWDM grid with a reach up to 80km)

65 sfpStatus (1.5)

65.1 Overview

Changelog

1.5

- Changed
 - sfpConstants from 1.3 to 1.4

65.2 Type Reference

65.2.1 SfpDiagnostics

Real Time Diagnostic for SFP/SFP+/QSFP28/QSFP+, with A2h/A0h memory space address in square brackets. (SFF 8472 Rev12.3 spec Table 9-11) (SFF 8636 Rev2.10 spec Table 6-1)

struct

temp	float Internally measured module temperature (°C). [96-97], [22-23]
vcc	float Internally measured supply voltage in transceiver (V). [98-99], [26-27]
txPwr	list of float Measured TX output power (mW). [102-103], [50-57]
rxPwr	list of float Measured RX input power (mW). [104-105], [34-41]

65.2.2 SfpStatus

SFP Status Data Fields, with A0h memory space address in square brackets. (SFF 8472 Rev12.3 spec Table 4-1) QSFP A0h memory space address is specified next to the SFP A0h. (SFF 8636 Rev 2.10 spec Table 6-14)

struct

identifier	sfpConstants.Identifier Type of transceiver (SFF 8024 Rev4.8a spec Table 4-1) [0],[128]
transceiver	list of sfpConstants.Transceiver Code for electronic or optical compatibility [3-10], [131-138]
brNominal	float Nominal signalling rate, units of 100 MBd. [12], [140/222]
vendorName	string SFP vendor name. [20-35], [148-163]
vendorOUI	string SFP vendor IEEE company ID. [37-39], [165-167]
vendorPN	string Part number provided by SFP vendor. [40-55], [168-183]
	string

vendorRev	Revision level for part number provided by vendor. [56-59], [184-185]
wavelength	string Laser wavelength. [60-61], [186-187]
vendorSN	string Serial number provided by SFP vendor. [68-83], [196-211]
dateCode	string Vendor's manufacturing date code. [84-91], [212-219]
diagnostics	optional SfpDiagnostics Optional Diagnostics Monitor Data.

66 slateInsertion (1.0)

66.1 Overview

```
# Changelog

## 1.0
- **Added**
  - First version of module
```

66.2 Type Reference

66.2.1 SlateInsertion

Configure slate insertion for a CoderService.

struct

trigger	SlateInsertionTrigger What events will trigger slate insertion
slateId	string A unique identifier for the slate

66.2.2 SlateInsertionTrigger

enum

DPI
FORCED

67 slotInterface (1.6)

67.1 Overview

- This file defines the external RPC interface to forward
- GetSlotInterface request to the correct card.

Changelog:

1.6

- Changed
 - ipInterface from 1.5 to 1.6

1.5

- Changed
 - ipInterface from 1.4 to 1.5

67.2 Command Reference

67.2.1 GetSlotInterfaces

- message `GetSlotInterfaces.Request`
- message `GetSlotInterfaces.Response`
- message `GetSlotInterfaces.Failure`

67.3 Type Reference

67.3.1 GetSlotInterfaces.Failure

empty `struct`

67.3.2 GetSlotInterfaces.Request

`IpInterfaceRequest`

67.3.3 GetSlotInterfaces.Response

`struct`

data `map` from `UUID` to `ipInterface.IpInterface`

67.3.4 IpInterfaceRequest

`struct`

slot `int`

68 st2110Status (1.4)

68.1 Overview

```
# Changelog
## 1.4
- **Changed**
  - `audioProfile` from 2.4 to 2.5

## 1.3
- **Changed**
  - `st2110Types` from 1.0 to 1.2.
  - `audioProfile` from 2.3 to 2.4
- **Removed**
  - `St2110AudioPacketTime` moved to `st2110Types` 1.2.
```

68.2 Type Reference

68.2.1 St2110AncStatus

struct

ipStreamUuid	UUID
standard	st2110Types.St2110Standard
sdpStatus	list of string

68.2.2 St2110AudioStatus

struct

ipStreamUuid	UUID
standard	st2110Types.St2110Standard
packetTime	st2110Types.St2110AudioPacketTime
totalChannels	int
sampleRate	optional audioProfile.SampleRate
sdpStatus	list of string

68.2.3 St2110Status

variant

videoStatus	St2110VideoStatus
audioStatus	St2110AudioStatus
ancStatus	St2110AncStatus

68.2.4 St2110VideoStatus

struct

ipStreamUuid	UUID
standard	st2110Types.St2110Standard

sdpStatus

list of string

69 st2110Types (1.2)

69.1 Overview

```
# Changelog

## 1.2
- **Added**
- `St2110AudioPacketTime`
```

69.2 Type Reference

69.2.1 St2110AudioPacketTime

Valid audio packet time according to SMPTE ST2110.

enum

N_A	Not applicable
US83_3	83.3 us
US125	125 us
MS1	1ms

69.2.2 St2110Standard

The SMPTE ST2110 standards that describe video, audio or ancillary content.

enum

ST2110_20	ST2110 uncompressed video standard
ST2110_22	ST2110 compressed video standard
ST2110_30	ST2110 audio standard
ST2110_31	ST2110 AES3 formatted audio
ST2110_40	ST2110 ancillary standard

70 streamSource (1.0)

70.1 Type Reference

70.1.1 StreamSource

StreamSource types of stream sources.
@param MAIN the main stream source.
@param BACKUP the backup stream source.

enum

MAIN

BACKUP

71 subtitling (1.1)

71.1 Overview

```
# Changelog

## 1.1
- **Changed**
- `component` from 1.3 to 1.5
```

71.2 Type Reference

71.2.1 SubtitleComponentPattern

Transport stream subtitle source.

struct

language	string Three letter language code.
component	optional component.SubtitleFormat Subtitle format.
pid	optional int Subtitle PID.

72 testGenComponent (1.1)

72.1 Overview

New in v1.1:
- Use version 1.1 of testGeneratorProfile that includes Freeze Frame
New in v1.0:
- These types were factored out from coderService

72.2 Type Reference

72.2.1 TestGenComponent

Configuration for a test pattern generator profile.
@param id UUID of the test pattern generator profile containing the configuration.
@param cpy A copy of the content of the test pattern generator profile.
@param enable Parameter to force test generation to override any input

struct

profile	TestGenComponent.profile
enable	bool

72.2.2 TestGenComponent.profile

struct

id	UUID
cpy	optional testGeneratorProfile.TestGeneratorProfile

73 testGenerator (1.1)

73.1 Overview

```
# Changelog

## 1.1
- **Changed**
  - Added Freeze Frame as a signal loss option
```

73.2 Type Reference

73.2.1 BitmapBackgroundProperty

Specifies the color property of the background in a bitmap.

enum

BLACK	Make the bitmap background solid black.
TRANSPARENT	Make the bitmap background transparent, such that the bitmap in essence becomes an overlay.

73.2.2 Color

A color defined in terms of RGB.

struct

red	int Amount of red. Value 0 to 255.
green	int Amount of green. Value 0 to 255.
blue	int Amount of blue. Value 0 to 255.

73.2.3 SignalLoss

The types of output that can be produced in the event of signal loss.

enum

COLOR_BAR	Displays color bar image.
BLACK_FRAME	Displays black frame.
FREEZE_FRAME	Displays last frame available.
SYNC_LOSS	Displays nothing.
TEST_GENERATOR	

73.2.4 TextPosition

Specify the vertical placement for the text.

enum

TOP	Place it at the top of the image.
MIDDLE	Place it in the middle of the image.
BOTTOM	Place it at the bottom of the image.

74 testGeneratorProfile (1.1)

74.1 Overview

New in v1.1:
- update to 1.1 version of testGenerator (includes Freeze Frame)

74.2 Command Reference

74.2.1 GetTestGeneratorProfiles

- message **GetTestGeneratorProfiles.Request**
- message **GetTestGeneratorProfiles.Response**
- message **GetTestGeneratorProfiles.Failure**

74.2.2 SetTestGeneratorProfiles

- message **SetTestGeneratorProfiles.Request**
- message **SetTestGeneratorProfiles.Response**
- message **SetTestGeneratorProfiles.Failure**

74.2.3 DeleteTestGeneratorProfiles

- message **DeleteTestGeneratorProfiles.Request**
- message **DeleteTestGeneratorProfiles.Response**
- message **DeleteTestGeneratorProfiles.Failure**

74.3 Type Reference

74.3.1 DeleteTestGeneratorProfiles.Failure

empty **struct**

74.3.2 DeleteTestGeneratorProfiles.Request

struct

ids **list of UUID**

74.3.3 DeleteTestGeneratorProfiles.Response

empty **struct**

74.3.4 GetTestGeneratorProfiles.Failure

empty **struct**

74.3.5 GetTestGeneratorProfiles.Request

empty **struct**

74.3.6 GetTestGeneratorProfiles.Response

struct

data	map from UUID to TestGeneratorProfile
info	GetTestGeneratorProfiles.Response.info

74.3.7 GetTestGeneratorProfiles.Response.info

struct

totalEntries	int
--------------	-----

74.3.8 MovingBox

Parameters for specifying properties of test generation moving box
 @param enable Enable moving box.
 @param color Enum describing color of moving box.

struct

enable	bool
color	testGenerator.Color

74.3.9 SetTestGeneratorProfiles.Failure

empty struct

74.3.10 SetTestGeneratorProfiles.Request

struct

data	map from UUID to TestGeneratorProfile
------	---------------------------------------

74.3.11 SetTestGeneratorProfiles.Response

empty struct

74.3.12 TestGenVariant

Variant for the different types of test generation.

variant

lipSync	TestGenVariant.lipSync variant specifies that the profile activates lipSync.
testGenerator	TestGenerator variant contains test generation parameters.

74.3.13 TestGenVariant.lipSync

empty **struct**

74.3.14 TestGenerator

Properties of test generation

struct

testPattern	testGenerator.SignalLoss Enum describing type of pattern to be generated.
testTone	bool Audio tone generation for channels.
text	Text Parameters regarding adding text to the test generation.
box	MovingBox Parameters regarding unsng a moving box for the test generation.

74.3.15 TestGeneratorProfile

TestGeneratorProfileStruct

74.3.16 TestGeneratorProfileStruct

top level profile structure

struct

label	string Name of the profile
enable	bool Parameter to force test generation to override any input.
testGenVariant	TestGenVariant either lipSync or test generator.

74.3.17 Text

Parameters for specifying properties of test generation text
 @param enable Enable text.
 @param position Specifies screen location for the text.
 @param backgroundType Specifies the color property of the background.

struct

enable	bool
position	testGenerator.TextPosition
backgroundType	testGenerator.BitmapBackgroundProperty

75 timeMode (1.0)

75.1 Type Reference

75.1.1 TimeMode

Specifies the source of timing information.

enum

NTP

PTP_IF_LOCKED

PTP_FORCED

76 timecodeConfig (1.0)

76.1 Type Reference

76.1.1 TimecodeSource

The clock source to be used by the encoder.

enum

OFF

SMPTE_12M

PTP

NTP

77 timecodeStatus (1.0)

77.1 Type Reference

77.1.1 Timecode

- Timecode information from SMPTE 12M, PTP or NTP source.

struct

timecode	UnixTime Current time in seconds in unix format
isPresent	bool Availability of timecode information

77.1.2 UnixTime

bigint

78 transceiverStatus (1.1)

78.1 Type Reference

78.1.1 CableLengthIndicator

Information about the cable length indicator measurement. The measurement is given in terms of the Belden 1694A cable.

struct

length	int An indication of the cable length carrying the SDI input signal [m].
lengthPrecision	int The precision of the cable length measurement [m]. The actual length of the cable is likely to be in the interval [length - precision, length + precision].
maxLength	int The maximum supported cable length [m].
threshold	int The threshold below which measurements of cable length are very imprecise [m].

78.1.2 DataRate

The various input data rates the transceiver can lock onto.

enum

DR_11_88_GBPS	
DR_5_94_GBPS	
DR_2_97_GBPS	
DR_1_485_GBPS	
DR_270_MBPS	
NO_LOCK	

78.1.3 SignalProperties

struct

dataRate	DataRate An indication of the data rate of the SDI signal that the transceiver has locked.
cableLength	optional CableLengthIndicator An indication of the cable length carrying the SDI input signal [m]. N/A if dataRate is NO_LOCK.
horizontalEyeOpening	optional float Horizontal eye opening of the equalized SDI signal [UI]. Not available if dataRate is NO_LOCK. Not available for SD.

78.1.4 TransceiverStatus

Status about the quality of an SDI signal. These measurements are taken by the transceiver which is connected to the physical connector carrying the SDI signal.

struct

carrierDetect	bool True if the transceiver has detected that there is an input signal from an SDI cable.
signalProperties	optional SignalProperties Properties of the input signal that are available if a carrier is detected.

79 tsDestination (1.0)

79.1 Overview

New in v1.0:

- These types were factored out from coderService
- Was TsOut before. Got renamed to TsDestination

79.2 Type Reference

79.2.1 TsDestination

struct

serviceName	string
serviceId	int
serviceProvider	string
pcrInterval	float
pcrPid	int
pmtPid	int

80 tsSource (1.4)

80.1 Overview

```
# Changelog
## 1.4
- **Added**
  - Amend `switchbackDelay` documentation in `InputRedundancy`; milliseconds, not seconds.
  - `switchDelay` to `InputRedundancy`

## 1.3
- **Added**
  - `FLOATING` to `SwitchingMode`.

## 1.2
- **Changed**
  - `subtitling` from 1.0 to 1.1
```

80.2 Type Reference

80.2.1 HotStandby

HotStandby configuration parameters.
 @param sourceProperty specifies whether the stream sources are identical or not.

struct

sourceProperty	SourceProperty
----------------	-----------------------

80.2.2 IdenticalSources

IdenticalSources configuration parameters.
 @param maxOffset The maximum tolerated delay between identical sources in ms.

struct

maxOffset	int
-----------	------------

80.2.3 InputRedundancy

InputRedundancy configuration parameters.
 @param mode the input redundancy switching mode.
 @param switchbackDelay a delay in milliseconds to wait before reverting to main. N/A for switching
 @param switchDelay A delay in milliseconds to wait before switching to non-active source. N/A for
 @param manualOverride a chosen preferred source will be forced even if it is not present.
 @param hotStandby the hot standby configuration parameters.

struct

mode	SwitchingMode
switchbackDelay	int
switchDelay	int
manualOverride	optional streamSource.StreamSource
hotStandby	HotStandby

80.2.4 SourceProperty

SourceProperty switching mode.
 @param nonIdenticalSources the stream sources are not identical.
 @param identicalSources identical stream sources configuration parameters.

variant

nonIdenticalSources	SourceProperty.nonIdenticalSources
identicalSources	IdenticalSources

80.2.5 SourceProperty.nonIdenticalSources

empty **struct**

80.2.6 SwitchingMode

InputRedundancy switching mode.

.

enum

REVERTING	allows stream source switching whenever the main is back. .
FLOATING	allows switching to another input and use it as long as it's error free.
OFF	Input Redundancy is turned off, no switching will takeplace.

80.2.7 TsSource

TS video source parameters.
 @param sources a map of the current stream source and their uuids. a MAIN stream source must always be present.
 @param inputRedundancy the input redundancy configuration parameters.
 @param subtitlePriorityList List of SubtitleComponentPattern describing which subtitle component is the most important.

struct

sources	map from streamSource.StreamSource to UUID
inputRedundancy	optional InputRedundancy
subtitlePriorityList	list of subtitling.SubtitleComponentPattern

81 vancProfile (1.5)

81.1 Overview

```
# Changelog:
## 1.5
- **Added**
  - enum `DpiSource`
  - member 'source' to Dpi

## 1.4
- **Added**
  - type `DvbSubtitle`
  - member `dvbSubtitle` to VancParams
  - value `DVBSUBTITLE` to VancParamsEnum
```

81.2 Command Reference

81.2.1 GetVancProfiles

- message **GetVancProfiles.Request**
- message **GetVancProfiles.Response**
- message **GetVancProfiles.Failure**

81.2.2 SetVancProfiles

- message **SetVancProfiles.Request**
- message **SetVancProfiles.Response**
- message **SetVancProfiles.Failure**

81.2.3 DeleteVancProfiles

- message **DeleteVancProfiles.Request**
- message **DeleteVancProfiles.Response**
- message **DeleteVancProfiles.Failure**

81.3 Type Reference

81.3.1 Anc

Anc.

struct

did	int vanc did.
sdid	int vanc sdid.
line	optional int line selection.

81.3.2 ClosedCaption

empty **struct**

81.3.3 DataSelection

Specifies which ancillary data packets are included in the SMPTE 2038 PID. An empty variant is treated as if auto was specified.

variant

manual

DataSelection.manual

Only ancillary data that matches the provided list of DID/SDID pairs is used (up to 4 pairs supported).

auto

DataSelection.auto

Use any incoming data.

81.3.4 DataSelection.auto

empty **struct**

81.3.5 DataSelection.manual

struct

input

list of **Anc**

81.3.6 DeleteVancProfiles.Failure

empty **struct**

81.3.7 DeleteVancProfiles.Request

struct

ids

list of **UUID**

81.3.8 DeleteVancProfiles.Response

empty **struct**

81.3.9 Dpi

Configuration of the DPI data

struct

source

DpiSource

Source of the DPI data

heartbeat

bool

Whether heartbeat is on or off

heartbeatInterval	int Interval between each heartbeat
ptsOffset	int SCTE-35 PTS adjustment
index	int SCTE 104 DPI PID index

81.3.10 DpiSource

enum

ANC
TCP

81.3.11 DvbSubtitle

DVB subtitle

struct

teletext **Teletext**
EBU teletext to be converted to DVB subtitles

81.3.12 EN301775

struct

teletext **optional Teletext**
vps **optional Vps**
wss **optional Wss**
cc **optional ClosedCaption**
dataId **optional int**

81.3.13 GetVancProfiles.Failure

empty **struct**

81.3.14 GetVancProfiles.Request

empty **struct**

81.3.15 GetVancProfiles.Response

struct

data **map** from **UUID** to **VancProfile**
info **GetVancProfiles.Response.info**

81.3.16 GetVancProfiles.Response.info

struct

totalEntries **int**

81.3.17 SetVancProfiles.Failure

empty **struct**

81.3.18 SetVancProfiles.Request

struct

data **map** from **UUID** to **VancProfile**

81.3.19 SetVancProfiles.Response

empty **struct**

81.3.20 Smpte2038

Smpte 2038.

struct

dataSelection	DataSelection modes of ANC data selection.
delay	int adds offset of x ms to PTS.
maxBitrate	optional int sets a max bitRate.

81.3.21 Teletext

struct

input	VancStandard
teletext	list of TeletextPsi
lineFilter	list of int

81.3.22 TeletextPsi

struct

page	int
language	string
teletextType	TeletextType

81.3.23 TeletextType

enum

START	
SUBTITLE	
HEARING_IMPAIRED	

81.3.24 VancParams

Configuration of the type of ancillary data in transport stream

variant

dpi	Dpi Digital program insertion messages according to scte-35
en301775	EN301775 VBI in DVB according to ETSI EN 301 775
smpte2038	Smpte2038 Ancillary data packets in a transport stream according to SMPTE ST 2038
dvbSubtitle	DvbSubtitle DVB subtitles according to ETSI EN 300 743

81.3.25 VancParamsEnum

enum

DPI	
EN301775	
SMPTE2038	
DVBSUBTITLE	

81.3.26 VancProfile

VancProfileStruct

81.3.27 VancProfileStruct

struct

label	string
vancParams	VancParams

81.3.28 VancStandard

enum

OP_47	
SMPTE_2031	
VBI	

81.3.29 Vps

empty **struct**

81.3.30 Wss

empty **struct**

82 vancStatus (1.11)

82.1 Overview

Changelog:

1.11

- **Changed**
 - sourceIdentifier member in dpiInput.

1.10

- **Changed**
 - dpiIn member in VancStatus from DpiInput to list(DpiInput).
 - Added timeSinceLastReceived to DpiInput.
 - Added lastGeneratedCommand to DpiInput.
 - Added DpiCommand type

1.9

- **Changed**
 - added value timeSinceLastReceived to DpiStatus
 - added value timeSinceLastGenerated to DpiStatus
 - added value lastCommandType to DpiStatus

1.8

- **Changed**
 - vancProfile from 1.4 to 1.5

1.7

- **Changed**
 - added value dvbSubtitle to VancComponentStatus
 - added value DVBSUBTITLESTATUS to VancComponentStatusEnum
 - changed function VancComponentStatus2Enum to accomodate DVBSUBTITLESTATUS
- **Added**
 - struct DvbSubtitleStatus

82.2 Type Reference

82.2.1 AspectRatioInput

struct

standard	aspectRatio.AspectRatioStandard
format	afd.AfdCode
is16x9	bool

82.2.2 DataIdentifier

The Data Identifier (DID) and Secondary Data Identifier (SDID) words which together indicate a type of ancillary data.

struct

did	int The Data Identifier word.
sdid	int The Secondary Data Identifier word.
lineNum	int The video line number of the DID/SDID.

82.2.3 DpiCommand

DPI Command info

struct

commandType	int Last SCTE command type.
descriptorIds	list of int List of segmentation descriptor IDs.

82.2.4 DpiInput

DPI input status.

struct

sourceIdentifier	optional int identifies the source of the input. Encoder: ANC is represented by -1, TCP input is represented by its identifier.
received	bool Received DPI input.
invalidPackets	int Input contained invalid packets.
pidIndexes	list of int
timeSinceLastReceived	int
lastGeneratedCommand	optional DpiCommand

82.2.5 DpiStatus

struct

heartbeat	bool
enabled	bool
running	bool
pidIndex	int
timeSinceLastReceived	int
timeSinceLastGenerated	int

lastCommandType	int
discardedPackets	int

82.2.6 DvbSubtitleStatus

DVB subtitle status

struct

enabled	bool Set to false if the feature is prevented from running (by licences, invalid config etc)
timeSinceLastReceived	int Seconds since last valid EBU subtitle input
timeSinceLastGenerated	int Seconds since last generated DVB subtitle

82.2.7 En301775Status

struct

ttxtStatus	optional TeletextStatus
------------	-------------------------

82.2.8 Smppte2038Status

Smppte 2038 Status.

struct

enabled	bool
running	bool
dataIdentifiers	list of DataIdentifier

82.2.9 TeletextInput

struct

input	vancProfile.VancStandard
linesIn	list of int
invalidPackets	int
received	bool

82.2.10 TeletextStatus

struct

standard	vancProfile.VancStandard
enabled	bool
running	bool
linesOut	list of int

82.2.11 VancComponentStatus

variant

dpi	DpiStatus
en301775	En301775Status
smpte2038	Smpte2038Status
dvbSubtitle	DvbSubtitleStatus

82.2.12 VancComponentStatusEnum

enum

DPISTATUS
En301775STATUS
SMPTE2038STATUS
DVBSUBTITLESTATUS

82.2.13 VancStatus

struct

txtIn	list of TeletextInput
dpiIn	list of DpiInput
arIn	list of AspectRatioInput
outputs	map from int to VancComponentStatus
sources	map from int to set of int
dataIdentifiers	list of DataIdentifier

83 videoProfile (2.9)

83.1 Overview

```
# Changelog:
## 2.9
- **Changed**
  - Bitrate changed from int to variant of int (cbr) and struct containing two ints (cvbr)
```

83.2 Command Reference

83.2.1 GetVideoProfiles

- message **GetVideoProfiles.Request**
- message **GetVideoProfiles.Response**
- message **GetVideoProfiles.Failure**

83.2.2 SetVideoProfiles

- message **SetVideoProfiles.Request**
- message **SetVideoProfiles.Response**
- message **SetVideoProfiles.Failure**

83.2.3 DeleteVideoProfiles

- message **DeleteVideoProfiles.Request**
- message **DeleteVideoProfiles.Response**
- message **DeleteVideoProfiles.Failure**

83.3 Type Reference

83.3.1 Afd

enum

Off
Passthrough

83.3.2 AspectRatio

enum

AR_TRANSPARENT
AR_16x9
AR_14x9
AR_4x3
AR_AFD
AR_WSS
AR_WSS18
AR_VideoIndex_11_324
AR_VideoIndex_8_321

83.3.3 AvcLevel

enum

AUTO
 LEVEL_3
 LEVEL_3p1
 LEVEL_3p2
 LEVEL_4
 LEVEL_4p1
 LEVEL_4p2
 LEVEL_5
 LEVEL_5p1

83.3.4 AvcParameters

struct

profile	AvcProfile
level	AvcLevel
codingMode	CodingMode
idrFrequency	int
cabac	bool

83.3.5 AvcProfile

enum

MAIN
 HIGH
 HIGH10
 HIGH42210
 CONSTRAINED_BASELINE

83.3.6 Bitrate

variant

cbr	int
cvbr	Cvbr

83.3.7 CodecParameters

variant

hevc	HevcParameters
avc	AvcParameters
mpeg2	Mpeg2Parameters

83.3.8 CodingMode

enum

AUTO
FRAME
FIELD
PAFF
MBAFF

83.3.9 Cvbr

struct

maxBitrate	int
averageBitrate	int

83.3.10 DeleteVideoProfiles.Failure

empty struct

83.3.11 DeleteVideoProfiles.Request

struct

ids	list of UUID
-----	--------------

83.3.12 DeleteVideoProfiles.Response

empty struct

83.3.13 GetVideoProfiles.Failure

empty struct

83.3.14 GetVideoProfiles.Request

empty struct

83.3.15 GetVideoProfiles.Response

struct

data	map from UUID to VideoProfile
info	GetVideoProfiles.Response.info

83.3.16 GetVideoProfiles.Response.info

struct

totalEntries	int
--------------	-----

83.3.17 GopMode

enum

Static	
Dynamic	

83.3.18 GopParameters

struct

size	int
gopMode	GopMode
structure	GopStructure
maxBframes	int
ldb	bool
hierarchical	bool

83.3.19 GopStructure

enum

I_ONLY	
IP	
IPB	
GDR	

83.3.20 HevcLevel

enum

AUTO	
LEVEL_3	
LEVEL_3p1	
LEVEL_4	
LEVEL_4p1	
LEVEL_5	
LEVEL_5p1	

83.3.21 HevcParameters

struct

profile	HevcProfile
level	HevcLevel
tier	HevcTier
idrFrequency	int

83.3.22 HevcProfile

enum

MAIN	
MAIN10	
MAIN42210	

83.3.23 HevcTier

enum

Main	
High	

83.3.24 Latency

enum

NORMAL	
LOW	
ULL	

83.3.25 Mpeg2Level

enum

MAIN	
HIGH	

83.3.26 Mpeg2Parameters

struct

profile	Mpeg2Profile
level	Mpeg2Level
codingMode	CodingMode
openGop	bool

83.3.27 Mpeg2Profile

enum

MAIN	
PROFILE_422P	

83.3.28 ProfileType

enum

ENCODER
DECODER
TRANSCODER

83.3.29 ScanningMode**enum**

INTERLACED
PROGRESSIVE

83.3.30 SetVideoProfiles.Failureempty **struct****83.3.31 SetVideoProfiles.Request****struct**

data	map from UUID to VideoProfile
------	--

83.3.32 SetVideoProfiles.Responseempty **struct****83.3.33 VideoCodec****enum**

HEVC
AVC
MPEG2
JPEG2000

83.3.34 VideoProfile**VideoProfileStruct****83.3.35 VideoProfileStruct****struct**

profileType	ProfileType
label	string
bitDepth	int
latency	Latency
resolution	VideoResolution
codec	VideoCodec
bitrate	Bitrate

cparams	CodecParameters
ar	AspectRatio
fallbackAr	AspectRatio
gop	GopParameters
afd	Afd
wssBlanking	bool

83.3.36 VideoResolution

struct

horizontal	rawVideo.HorizontalResolution
vertical	rawVideo.VerticalResolution
chromaSampling	rawVideo.ChromaSampling
scan	ScanningMode
fps	rawVideo.FrameRate
followInput	bool

84 videoStatus (2.12)

84.1 Overview

Video status types and objects. This IDCL is part of the IDCL sent to the GUI

Changelog:

2.12

- **Changed**
 - videoProfile from 2.8 to 2.9
 - ccPresent added to Es

2.11

- **Changed**
 - Ts.bitRate changed from int to bigint to allow for high bitrates, which will be necessary for UHD

84.2 Type Reference

84.2.1 Es

Status for coded video.

struct

valid	bool
hSize	int
vSize	int
bitRate	int
cpb	optional int
bitDepth	int
chroma	int
codec	videoProfile.VideoCodec
aspectRatio	optional videoProfile.AspectRatio
framerate	optional rawVideo.FrameRate
scan	optional videoProfile.ScanningMode
colorDescPresent	bool
colorPrimaries	colorProfile.ColorSpace
transferChar	colorProfile.TransferFunc
matrixCoeffs	colorProfile.ColorSpace
idrOnScte35	optional int
ccPresent	bool

84.2.2 InputStatus

struct

id	InstanceId
input	VideoFormat

84.2.3 InstanceId

int

84.2.4 OutputStatus

struct

id	InstanceId
output	VideoFormat

84.2.5 Raw

struct

valid	bool
numFrames	int
videoMode	optional VideoModes
framerate	optional rawVideo.FrameRate
chroma	optional int
bitDepth	optional int
vitcPresent	bool
vitcLineNr	optional int
ccPresent	bool
timecode	optional timecodeStatus.Timecode

84.2.6 Ts

struct

bitRate	bigint
ccError	int
pcrJitterError	optional bool
pcrDisc	optional bool
pcrIntervalErr	optional bool
es	Es
pid	optional int

84.2.7 VideoFormat

variant

raw	Raw
es	Es
ts	Ts

84.2.8 VideoModes

enum

V_UNKNOWN

V_480i

V_576i

V_720p

V_1080i

V_1080p

V_2160p

84.2.9 VideoStatus

struct

input

InputStatus

output

OutputStatus